



# Development of a novel macrostructured cathode for large-area neutron detectors based on the $^{10}\text{B}$ -containing solid converter



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## ARTICLE INFO

### Article history:

Received 26 October 2012

Received in revised form

8 May 2013

Accepted 3 June 2013

Available online 18 June 2013

### Keywords:

Neutron detector

Boron converter

GEANT4

GARFIELD

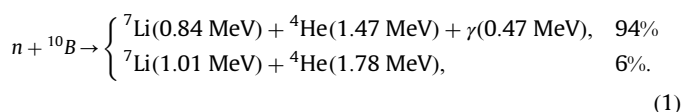
## ABSTRACT

We present a novel design for a macrostructured cathode that can be coated with a thin layer of the  $^{10}\text{B}$  solid converter and mounted to replace the Boron-lined flat parallel plates of a proportional counter used for slow neutron detection. The proposed design consists of a 3D regular pattern exhibiting millimeter deep grooves with an opening angle of  $\alpha = 45^\circ$ , which could be created in the substrate material by milling or forming. When a commonly used coating method like magnetron sputtering is employed to deposit the Boron-layer, due to the line-of-sight distribution of the ions, the thickness of the coating on the side of the grooves will be reduced by a factor  $\sim \sin \alpha/2$  with respect to the thickness of the layer deposited on a flat surface normal to the ion flux. The effective neutron absorption film thickness is in this case similar for the sidewalls of the grooves and a surface at normal incidence, yielding comparable absorption efficiencies. However, the escape efficiency for the reaction products is higher for the sidewalls, owing to the thinner coating. This leads to a higher overall detection efficiency for the grooved cathode when compared to a flat cathode with the same surface area and coated with a Boron layer with roughly the same thickness. In this paper we present and discuss the GEANT4 simulations performed to optimize the geometry of the cathode, the manufacturing and coating by magnetron sputtering, as well as the proof-of-principle measurements carried out in order to assess the performance of the proposed design.

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## 1. Introduction

In the few years since the  $^3\text{He}$  crisis has started [1], several detector designs based on  $^{10}\text{B}$ -containing solid converter have been proposed as alternative technologies for neutron scattering applications that require large-area neutron detectors [2–8]. The preference for this isotope is not surprising, as  $^{10}\text{B}$  has been the material of choice for slow neutron detection before the  $^3\text{He}$  gas became available in large quantities [9]. In a  $^{10}\text{B}$ -based neutron detector, the electrical signal is provided by the  $^7\text{Li}$ - and  $^4\text{He}$ -charged particles produced in the  $n+^{10}\text{B}$  reaction via two distinct branches:



In the detector systems described in Refs. [2–8], the  $^{10}\text{B}$  solid converter has the form of a thin layer coated either on the inner surface of the walls, or on flat plates playing the role of cathodes in gas-filled proportional counters. The reaction products have to escape the layer and reach the gas phase in order to be detected, therefore the thickness of the layer is a crucial parameter for the neutron detection efficiency.

The optimum thickness of a single  $^{10}\text{B}$  layer to be used as a converter for slow neutrons is dictated by the ranges of the reaction products in Boron, which are around  $3.5 \mu\text{m}$  for the 1.47-MeV alpha particle and  $\sim 2 \mu\text{m}$  for the 0.84-MeV Li ion [10]. Calculations indicate that a thickness around  $3 \mu\text{m}$  is the best compromise between the neutron absorption probability and the probability for the reaction products to escape the converter and produce a detectable signal [10,11]. This limits the efficiency of a detector incorporating one  $^{10}\text{B}$  layer to  $\sim 5\%$  for thermal neutrons ( $\lambda = 1.8 \text{ \AA}$ ), a value that is far below the 90% efficiency that could be achieved with a proportional tube filled with 8 bar of  $^3\text{He}$  gas.

Neutron detectors based on the  $^{10}\text{B}$  solid converter can reach detection efficiencies above 50% when several thin Boron layers

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are stacked together in a common detector housing, or many individual detector units, each incorporating a single Boron-layer, are closely packed together. Calculations for Boron layers deposited on substrates with flat surfaces indicate that over 20 substrates coated with Boron on both sides are required in order for the detector to achieve an efficiency around 50% for thermal neutrons [10]. The optimum thickness of each individual layer decreases with increasing the total number of layers in the stack, in order to minimize the absorption of the neutron beam in the Boron converter, while maximizing the escape probability. Thus, the replacement of the  $^3\text{He}$ -tubes with the technology based on the  $^{10}\text{B}$  solid converter requires dealing with numerous large-area layers, each of which must provide position information and electronic integration. For example, new instruments at the future high-flux neutron scattering facility European Spallation Source ESS AB at Lund, Sweden, require neutron detectors with an efficiency over 50% for thermal neutrons, position resolution in the millimeter range and the capability to handle more than  $10^6$  counts/s [12,13]. The tasks of designing and manufacturing a neutron detector that fulfills these requirements become very challenging. The production of a large number of closed coated units or flat Boron layers must be made in a way that is technically feasible and economically viable. Also, the large number of readout channels demands finding a strategy for the front-end electronics that ensures high signal quality at low cost per channel [2–5,7].

The performance of a  $^{10}\text{B}$ -based neutron detector is determined by the quality of the coating (uniformity, purity, thickness), which is crucial for the conversion efficiency, and the geometry of the substrate upon which the thin  $^{10}\text{B}$ -layer is coated. In principle, the substrate can be made in a variety of shapes (tubes, flat, box, etc.), provided that the coating can be uniformly applied on the side facing the gas, and that the electric field applied to multiply and collect the charge is not very much disturbed by the geometry itself, so that the charge collection efficiency is close to 100% and it does not significantly distort the time resolution of the detected neutron.

Geometrically, by inclining the Boron layer at an angle  $\alpha$  with respect to the direction of the incoming neutron beam, the conversion efficiency increases because the effective neutron absorbing film thickness is increased by a factor proportional to  $1/\sin \alpha$ , and the escape probability for the charged particles becomes larger since the reaction has higher chances to occur at lower depth with respect to the gas surface [14]. This approach is being used for the design of the *Jalousie* detector prototype, which consists of several lamellae coated with  $1\text{ }\mu\text{m}$  of  $^{10}\text{B}$  and mounted at  $10^\circ$  with respect to the incoming beam direction [5]. Based on the same design concept is the neutron detector currently under development at the Institute for Material Research Geesthacht, Germany, which consists of thin conversion Boron layers arranged at  $2\text{--}5^\circ$  with respect to the beam direction [6]. The main challenge that these detector types have to face is associated with the much larger surface area of  $^{10}\text{B}$  converters that is required to cover the desired solid angle, which implicitly leads to a higher number of detector channels than that needed by neutron detectors incorporating Boron layers orthogonal to the beam direction.

Another way to make use of the incidence angle in order to increase the efficiency of a single Boron layer is to create in the substrate material a 3D regular pattern like, e.g., inverted pyramids or trenches, with the sidewalls inclined at an angle  $\alpha$  with respect to the normal and a depth of several millimeters. After the patterning of the surface, the substrate material is coated with approximately the same amount of Boron that is required to coat a flat surface of the same area. When a commonly used coating method like magnetron sputtering is employed to deposit the Boron layer, due to the line-of-sight distribution of the ions, the thickness of the coating on the sidewalls will be reduced by a factor  $\sim \sin \alpha/2$  with respect to the thickness of the layer deposited on a flat surface normal to the ion

flux. In this case, the effective neutron absorption film thickness will be roughly the same for both surfaces, yielding similar neutron conversion efficiencies. However, the sidewalls of the pyramids or trenches are expected to have a higher overall efficiency because the neutron interaction with the Boron coating takes place at a smaller depth with respect to the gas surface. This leads to a reduced transit distance in the layer, which results in a larger escape probability for the charged ions. Thus, a neutron detector consisting of a Multi Wire Proportional Chamber in which the cathode plates are patterned and coated with Boron is expected to have an overall efficiency superior to one incorporating Boron-lined flat parallel plates.

A similar approach based on the patterning of the substrate material has been proposed in order to increase the efficiency of solid state neutron detectors [15]. In that case, the increase in efficiency was achieved by etching several micrometers deep dips or pores in the Silicon substrate, which were subsequently filled with either  $^6\text{Li}$ - or  $^{10}\text{B}$ -converter materials. The microstructures created in the substrate helped increasing the surface between the neutron converter and detector, while keeping a high probability for secondary particle detection, which yielded a higher detection efficiency when compared to a planar device of the same area [15].

In a recent work by Lacy and collaborators, a new design for the straw tubes consisting of star-shaped straws is investigated and discussed [16]. With the corrugated straw structures an increase of the boron-coated area by a factor of  $\sim 2$  can be achieved, which leads to an efficiency superior to that of a round tube with the same cross-sectional area. When the optimal coating thickness is employed, the efficiency of a stack of 31 star-shaped tubes is up to 40% higher than the efficiency of a stack populated with the same number of round tubes. The star-shapes tubes are easy to manufacture and the results of tests in strong gamma fields indicate low sensitivity for gamma radiation. The novel design will prove highly beneficial for applications requiring a large number of closely packed detectors.

In this paper, we report on the results of the development work of a cathode exhibiting a V-shaped regular pattern that can be coated with the  $^{10}\text{B}$  converter and used in large-area detectors for slow neutrons. We used the GEANT4 code to study the performance of such a design as a function of the height of the groove, by assuming two coating scenarios for the sidewalls. The calculations indicated that a cathode with  $45^\circ$  grooves coated with  $\sim 3\text{ }\mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  can enhance the sensitivity to slow neutrons with up to 40% when compared to a flat surface. Several cathodes with grooves of various heights were manufactured, coated with Boron by employing magnetron sputtering and used for proof-of-principle measurements. The conformality of the coating, as well as the influence of the 3D pattern on the electric field applied for charge amplification and collection were investigated with a  $4.7\text{ }\text{\AA}$  collimated neutron beam delivered by the FRM-II München research reactor. The experimental pulse-height spectra and efficiencies obtained in the measurements with the grooved cathodes were compared to the results of similar investigations carried out with a flat cathode coated with a Boron layer of approximately the same thickness. All results were compared to the predictions of the GEANT4 model, which proved to be able to provide an accurate description of the observed properties, although important processes like the generation of the low-energy electrons along the ion tracks and diffusion in the presence of an electric field were not included in the model.

## 2. Theoretical investigation of the grooved cathodes

Several criteria were considered when designing the 3D pattern to be created in the substrate material. In order to achieve a

high conversion efficiency, the pattern has to include grooves with sidewalls sufficiently large to increase the neutron interaction probability and also allow for an efficient charge collection. The shape of the groove has to be chosen such that the coating can be deposited as uniformly as possible on its surface. Also, the selected design of the cathode has to allow for an easy and fast production at low costs, as well as an easy assembly, handling and testing.

The pattern shown in Fig. 1 consisting of V-shaped grooves horizontally extending in one direction and with a depth of several millimeters was found to fulfill all the above requirements. Therefore, this pattern was chosen for detailed investigations such as optimization of the geometry in GEANT4, fabrication and testing with neutron beam.

### 2.1. Modeling the grooved cathode with the GEANT4 Monte Carlo toolkit

The optimization of the geometry of the grooved cathode was performed with the GEANT4 toolkit, which is widely used by the scientific community to simulate the passage of particles through matter and model complex radiation detectors [17]. The GEANT4 version used for the simulations presented in this work was the 4.9.3 release. The geometry of interest was built by coding in C++ language using combinations of simple shapes available directly as solid objects according to the CSG (Constructed Solid Geometry) specifications, or solids obtained by using Boolean operations like unions, intersections and subtractions. The monoenergetic incident neutrons were released from the built-in particle generator.

The physics processes relevant to the transport and interaction of the low-energy neutrons with the matter, as well as the electromagnetic and stopping physics relevant to the secondary charged particles, were accounted for by the reference list QGSP-BIC-HP [18]. This is one of the standard physics lists provided by GEANT4 with the source code, which consists of a set of consistent physics models for each particle in the application. The simulation was performed on an event-by-event basis and each primary particle (i.e., the neutron) was tracked from its release to the interaction with the Boron coating. The secondary charged particles produced in the  $n+^{10}\text{B}$  reaction of interest were tracked down until they deposited the entire kinetic energy and arrived to rest. In the geometry considered in this work, the media that could be traversed by the  $^7\text{Li}$  and  $^4\text{He}$  ions were the Aluminium substrate material, the Boron layer and the counting gas. In any of these media the ions travel along a straight path and undergo a continuous slowing down, thereby losing energy as a function of the specific stopping power of the material which they pass. Since only the ions that could reach the counting gas have a non-zero detection probability, the energy deposited in gas was the key parameter in assessing the performance of the investigated design.

The efficiency of the proposed design was investigated by recording the energy deposition spectra and taking the ratio between the total number of  $^7\text{Li}$  or  $\alpha$  ions detected in gas and the total number of incident neutrons impinging upon the cathode. The result represents a measure for the conversion efficiency only,  $\epsilon_{\text{conversion}}$ , not the total detection efficiency,  $\epsilon_{\text{det}}$ ,

which also depends on the ability to collect the charge generated in the gas volume, i.e.,  $\epsilon_{\text{det}} = \epsilon_{\text{conversion}} \times \epsilon_{\text{collection}}$ .

The parameters that describe the surface with inverted pyramids depicted in Fig. 1 are the length of the flat top  $f$ , the height  $h$  and the opening angle  $\alpha$  of the groove, and the thickness  $l$  of the substrate at the flat top (bottom). As it will be discussed in more detail in the next subsection, milling was identified as a fast and cheap production method for the grooved cathodes. Since the  $45^\circ$  milling cutters are already available commercially, we chose this value for the opening angle  $\alpha$  of the grooves in our design. The GEANT4 optimization process consisted in varying the  $f$ ,  $h$ , and  $l$  shape parameters in order to find the design with the highest conversion efficiency.

The GEANT4 arrangement used in the present simulations to investigate the performance of a  $10 \times 10 \text{ cm}^2$  cathode with  $45^\circ$  grooves is shown in Fig. 2. One surface of the grooved cathode made of Aluminium was positioned in direct contact with a thin  $^{10}\text{B}_4\text{C}$  layer. The assembly was placed inside of a detector housing made of Aluminium and filled with an  $\text{Ar}/\text{CO}_2$  (70/30) gas at room temperature. For this gas mixture at normal conditions, the predicted range of the 1.47- and 1.78-MeV alpha particles is  $\sim 7$  and 9 mm, respectively, and that of the 0.84- and 1.0-MeV  $^7\text{Li}$  ions is  $\sim 4$  mm.

The detector setup was irradiated from aside with an uniformly distributed neutron beam such that every neutron emitted by the particle generator struck the surface of the cathode. The incident neutrons first scattered through the Aluminium substrate and then interacted with the Boron coating, where the  $^{10}\text{B}(n,\alpha)^7\text{Li}$  reaction took place at a rate that depended on the energy of the incident neutrons, thickness of the coating, and abundance of the  $^{10}\text{B}$  isotope in the coating. The spectrum corresponding to the portion of the kinetic energy deposited in the counting gas by the  $^7\text{Li}$  and  $^4\text{He}$  ions was recorded for each run. The calculations were performed in forward irradiation geometry, meaning that the incident neutrons and the reaction products that contributed to the signal moved in the same direction.

One important issue about the proposed design concerns the conformality of the Boron coating on the sidewalls of the grooves. Since the chosen pattern consists of combinations of small flat

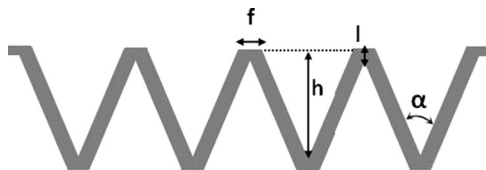


Fig. 1. Design proposed here for a cathode to be used in a  $^{10}\text{B}$ -based large-area neutron detector.

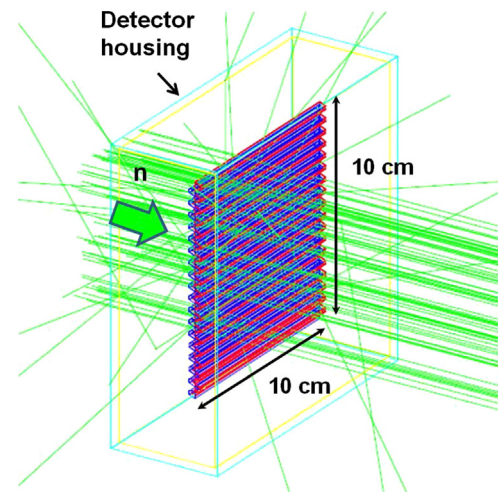


Fig. 2. Visualization of the GEANT4 setup used in the present work to optimize the design of the grooved cathode. The frame of the Aluminium-substrate material is represented in blue, the Boron coating is shown in red. The detector housing and counting gas are represented in “wireframe” mode with light blue and yellow colors, respectively. The green lines represent the trajectories of the incident, transmitted, and scattered neutrons. The direction of the incident beam is represented in the figure with the full green arrow. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

portions at  $90^\circ$  and portions at  $22.5^\circ$  with respect to the normal, it is of crucial importance that one finds a coating technique that can provide a uniform coverage of such a highly structured surface. Common methods like electron-beam evaporation or RF/DC magnetron sputtering are known for their directional nature or “line-of-sight” deposition, which leads to a great deal of self-shadowing that could cause a poor coverage of the surfaces that are not normal to the direction of the incident atom flux. Thus, when either of these two techniques is employed to coat the grooved cathode, only the flat top is expected to receive a uniform layer with the desired thickness, denoted here as  $t_{top}$ . In the present calculations, two different assumptions were made for the conformality of the coating on the sidewalls. The first scenario corresponds to the ideal case in which the coating is uniformly deposited on the sidewalls of the grooves, see Fig. 3a. The thickness of the Boron layer on the sidewalls, defined here as the orthogonal distance between the two surfaces of the layer and denoted as  $t_{side}$ , is then given by the relation

$$t_{side} = t_{top} \cdot \sin(\alpha/2), \quad (2)$$

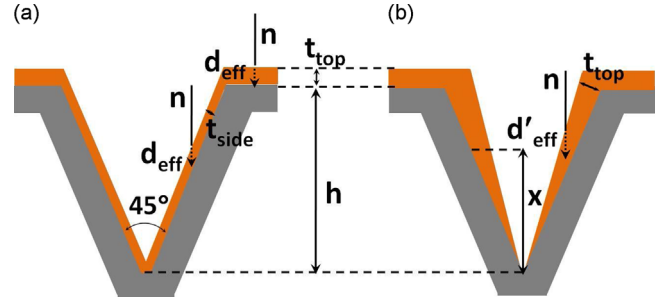
where  $\alpha$  is the opening angle of the groove. With this assumption, the effective neutron absorption film thickness,  $d_{eff}$ , is the same everywhere on the surface of the groove. In the second scenario, the converter is preferentially deposited on the upper half of the sidewalls. The coating was parameterized such that the thickness equals  $t_{top}$  only at the upper corner of the groove, decreases with the depth and reaches zero at the bottom, see in Fig. 3b. The variation of the thickness  $t_x$  with the distance  $x$  from the bottom of the groove is given by

$$t_x = x \cdot t_{top}/h, \quad x \leq h. \quad (3)$$

In this case, the increase in efficiency with respect to a flat cathode of the same active area arises from large absorption probabilities for the neutrons striking the upper half of the groove and large escape probabilities when the interaction takes place near bottom.

Fig. 4 shows the trend of the total number of ions recorded in the energy deposition spectra for  $10 \times 10 \text{ cm}^2$  grooved cathodes as a function of the height of the groove  $h$ , along with the similar integral obtained for a flat cathode of the same size. The calculations were performed for a boron layer of enriched  $^{10}\text{B}_4\text{C}$  with a density of  $2.34 \text{ g/cm}^3$  [24]. The thickness of the layer on the flat cathode and the flat portions of the grooved cathode was  $3 \mu\text{m}$ , and as given by Eqs. (2) and (3) for the sidewalls. Neutrons with an energy of  $3.63 \text{ MeV}$  ( $4.7 \text{ \AA}$ ) struck the cathodes at normal incidence.  $10^5$  neutrons were simulated for each run, which ensured adequate statistics in the histograms for a reasonable runtime. In the simulations, the height of the groove  $h$  was varied between  $0.25$  and  $5 \text{ mm}$ , and the length of the flat top  $f$  was chosen such that  $h/f \sim 3$ . All calculations were made for cathodes with a size of  $10 \times 10 \text{ cm}^2$ , therefore the increase of the size of the grooves from  $h=0.25 \text{ mm}$  to  $h=5 \text{ mm}$  led to a decrease of the number of grooves per surface area from  $\sim 450$  down to  $18$ , respectively. The results presented in the top panel of Fig. 4 were obtained by simply taking the integral of the whole calculated energy deposition spectrum, while the areas shown in the middle panel of the figure were calculated by considering only the ions that escaped the Boron layer with kinetic energies higher than the threshold set at  $150 \text{ keV}$ , which would be typically applied in practice in order to discriminate between the valid neutron events and  $\gamma$ -background.

The results of the simulations shown in the top panel of the figure indicate that indeed, the 3D pattern created in the substrate material leads to an increase of the total number of reaction products that reaches the gas phase, which, for small grooves with heights below  $1 \text{ mm}$ , can exceed the number of ions escaping from a thin flat layer of the same area by more than  $60\%$ . One can notice

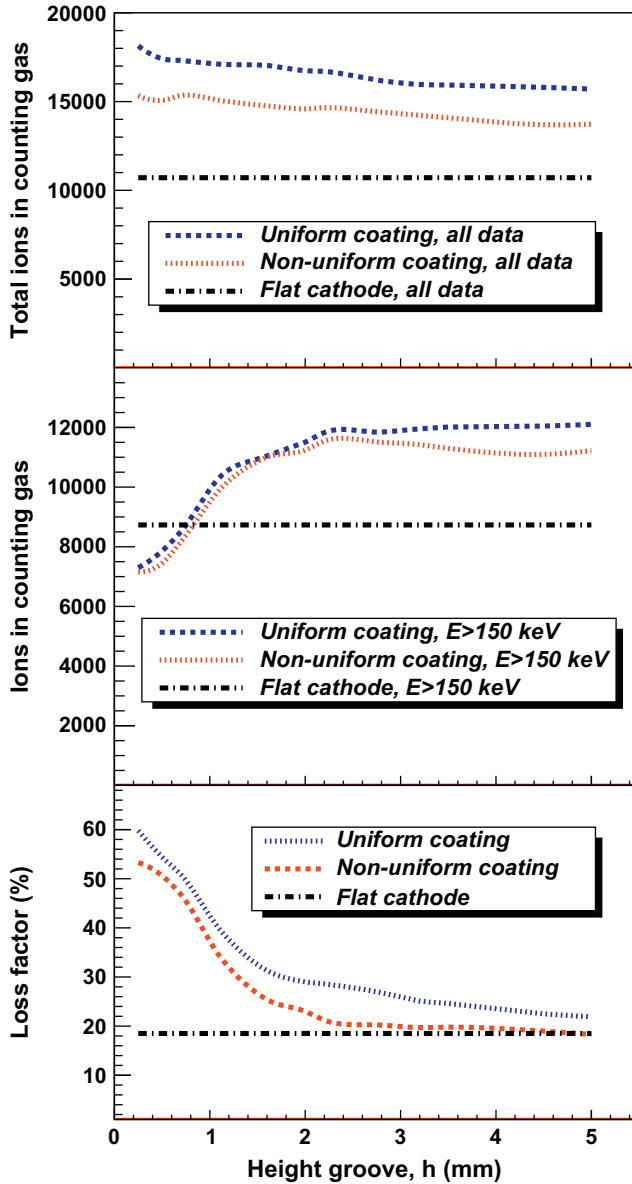


**Fig. 3.** Schematic representation of the Boron coating assumed in the GEANT4 simulations. The Aluminium-substrate is represented in grey, and the Boron coating is drawn in orange. (a) Uniform coating is assumed. The thickness of the Boron coating on the sidewalls of the groove, denoted as  $t_{side}$ , is related to the thickness of the coating on the flat top,  $t_{top}$ , by  $t_{side} = t_{top} \cdot \sin(\alpha/2)$ . The effective neutron absorption film thickness,  $d_{eff}$ , is the same everywhere on the surface of the groove. (b) Non-uniform coating is assumed. The thickness of the coating is  $t_{top}$  at the upper corner of the sidewalls, decreases with depth and reaches zero at the bottom of the groove. Note that for the clarity of the graphical presentation, the thickness of the coating is highly exaggerated. In practice, the thickness of the substrate material is two orders of magnitude larger than that of the Boron converter deposited on its surface. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

from the comparison of the results obtained for the two coating scenarios that the ideal, uniform coating of the sidewalls gives better results than the non-uniform coating. The evenly distributed converter material in the first case leads to integrals that are up to  $10\%$  higher, see top panel of Fig. 4. The simulation predicts a smooth decrease in the number of ions with increasing the height  $h$  for both coating scenarios.

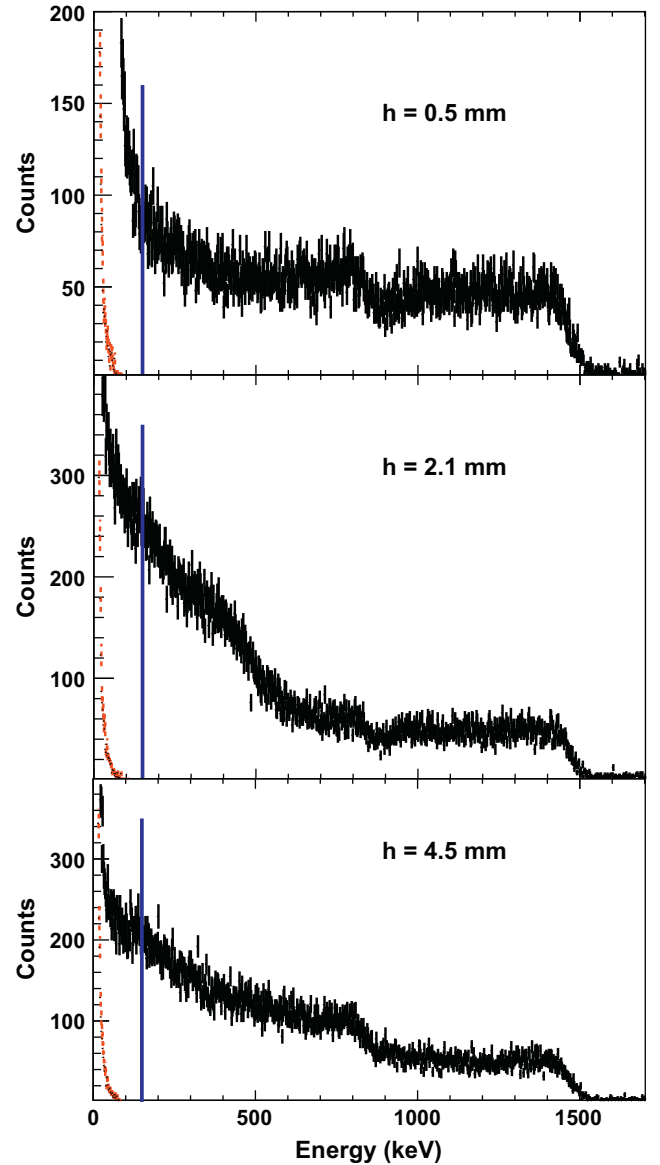
The trend shown by the full integrals of the energy deposition spectra undergoes a rather drastic change in appearance when a cut-off energy is applied, see middle panel of Fig. 4. When only the reaction products with an energy above  $150 \text{ keV}$  are considered in the assessment of the performance of the grooved cathode, up to  $60\%$  of the ions are lost in the case of the small grooves ( $h < 1 \text{ mm}$ ), see bottom panel of the same figure. The number of ions that slip below the energy threshold decreases rapidly with increasing the height of the groove to  $h \approx 2 \text{ mm}$ . According to the calculations,  $70\text{--}80\%$  of the total number of ions that escape the converter deposited on the surface of a cathode exhibiting grooves with heights larger than  $2 \text{ mm}$  have energies above  $150 \text{ keV}$ . This leads to a sensitivity for slow neutrons that can be with up to  $40\%$  higher than what it could be achieved with a flat cathode of the same active area and coated with a Boron layer of roughly the same thickness.

The trend of the curves shown in Fig. 4 could be understood by analyzing the corresponding simulated energy deposition spectra. The spectrum displayed in the top panel of Fig. 5 corresponds to the cathode with  $0.5 \text{ mm}$  deep grooves, the spectrum in the middle was collected for a cathode with  $2.1 \text{ mm}$  deep grooves and that showed in the bottom panel of the figure was obtained for the cathode with  $4.5 \text{ mm}$  deep grooves. The red spectra correspond to the  $\gamma$ -background calculated for each investigated design. In a real experiment, part of the  $\gamma$ -background is expected to arise from low-energy electrons generated in the photoelectric and Compton interaction of the secondary  $\gamma$ -rays of  $0.47 \text{ MeV}$ , emitted in the  $(n, ^{10}\text{B})$  reaction, with the aluminum-substrate or detector housing. Another contribution to the background comes in practice from the environmental  $\gamma$ -rays (cosmic-rays, natural radioactivity, various beamline components). The accurate modeling of these events requires the precise knowledge of the specific spectra, which are usually rather complex. In the present work, a simplified approach has been used to calculate the contribution to the background from the environmental  $\gamma$ -rays. As in the present study one is interested in the amount of energy deposited in the



**Fig. 4.** Top: simulated total number of charged ions that escapes the Boron converter and reach the gas phase as a function of the height of the groove, assuming uniform (dashed blue line) and non-uniform coating of the sidewalls (dotted red line), compared to a flat cathode (dash-dot black line). Middle: same as above, but with an energy threshold set at  $E = 150$  keV. Bottom: loss in efficiency caused by the energy threshold set at  $E = 150$  keV. All simulations were done for cathodes with a size of  $10 \times 10$  cm<sup>2</sup> and coated enriched <sup>10</sup>B<sub>4</sub>C with a density of 2.34 g/cm<sup>3</sup>. The thickness of the coating on the flat cathode and flat portions of the grooved cathode was 3  $\mu$ m and as given by Eqs. (2) and (3) for the sidewalls, respectively. The simulations were performed for  $10^5$  incident neutrons striking the cathode at normal incidence.

counting gas by the photoelectrons rather than their absolute number, the  $\gamma$ -background was calculated by irradiating the detector with a mixed beam of 4.7 Å neutrons and 5 MeV  $\gamma$ -rays. The result of the simulation of the  $\gamma$ -background for the small assembly depicted in Fig. 2 suggested that the energy deposited by the photoelectrons in the ArCO<sub>2</sub> gas ( $dE/dx \sim 20$  keV/cm for  $E_e < 5$  MeV, [19]) is less than 100 keV, see spectra drawn in red in Fig. 5. The amount of deposited charge is proportional to the path traveled in the counting gas by the photoelectrons, which depends on the geometry and size of the detector. In the small size setup used in the present simulations, chosen to resemble closely the test detector described in Section 5.2, this range was limited by



**Fig. 5.** Simulated energy deposition spectra for grooves with heights  $h = 0.5$  mm (top), 2.1 mm (middle) and 4.5 mm (bottom). The spectra include the energy deposited in gas by the reaction products ( $\alpha$  and <sup>7</sup>Li) and the photoelectrons produced by the background  $\gamma$ -rays, see text for details. The energy deposited in gas by the photoelectrons is also drawn separately in red. The blue vertical line corresponds to the energy cut-off set at 150 keV. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

the distance between the two parallel cathodes, which was  $\sim 8$  mm. The calculations performed for the groove heights selected for the study shown in Fig. 5 suggested that the number of photoelectrons reaching the gas phase increases with increasing the height of the groove. The larger groove size requires a thicker Aluminium-substrate, which leads to a greater probability for the photoelectric and Compton interaction of the background  $\gamma$ -rays. However, since the active area of the detector remains unchanged, the geometry of the groove plays an insignificant role for the amount of energy deposited in gas by the photoelectrons. Although the calculation suggests that a threshold at 100 keV is sufficient to exclude the  $\gamma$ -unwanted events for all investigated samples, experimental measurements indicated that a discriminator threshold set at 150 keV ensures a much better  $\gamma$ -rejection, see next sections. Therefore all efficiency values (theoretical and

experimental) reported in this work were evaluated with the  $n$ - $\gamma$  threshold set at  $E=150$  keV, which is represented in Fig. 5 by the blue vertical line.

The spectra shown in Fig. 5 were calculated for the coating scenario depicted in Fig. 3b. As it can be seen from the comparison of the top and the middle panel spectra, the simulated distributions change appearance when the groove height is increased from 0.5 to 2.1 mm. The spectrum calculated for the cathode with small grooves (Fig. 5, top) has the bulk of events at low energies and the remainder evenly distributed between 200 and 1500 keV. The flat portion above 200 keV shows the characteristics of a spectrum obtained for a planar Boron layer and arises from the interaction of the incident neutrons with the converter material deposited on the flat tops of the grooves. The low-energy peak centered around 50 keV originates from the charged particles released in the interaction of the neutrons with the Boron converter deposited on the sides of the grooves. The line-of-sight of the particles emitted from one sidewall of the groove is restricted by the opposite sidewall, which for the small grooves is located at a distance of several hundred of microns. This distance in gas is insufficient for the charged ions to deposit enough energy and generate a large signal before implanting into the opposite sidewall. This problem could be overcome either by increasing the size of the groove or by raising the pressure of the counting gas to values above 5 bar, which will reduce the range of the reaction products, and hence increase the energy deposition in the gas. The latter solution would require a more technically challenging design of the detector, which will then need to withstand large gas pressures.

It can be seen in the middle and the bottom panels of Fig. 5 that the increase of the size of the groove produces a smearing out of the sharp peak below 200 keV in the spectrum calculated for the cathode with 0.5 mm deep grooves. The flying path in the counting gas is now large enough for the charged particles generated on the sidewalls of the grooves to create a signal of several hundreds of keV. Increasing further the size of the groove does not change significantly the spectral features, which explains why the integrals plotted in Fig. 4 do not change much for grooves with heights larger than 2 mm. However, since the majority of events has energies below 500 keV, the efficiency of the grooved cathode is expected to be very sensitive to the energy cut-off.

The dependence of the conversion efficiency on the thickness of the Boron-layer for a cathode with 2.1 mm deep grooves is shown in Fig. 6. The calculations presented in the top panel of the figure were performed in forward-irradiation geometry (Boron-layer coated on the side of the cathode opposite to the neutron source), those illustrated in the bottom panel were carried out in backward-irradiation geometry (Boron-layer coated on the side of the cathode facing the source) for  $10^5$  incident neutrons with a wavelength of 4.7 Å striking the cathode orthogonal to the surface. The efficiency curves calculated for a flat cathode are also represented for comparison purposes. The efficiencies were extracted by dividing the number of detected ions with energies larger than 150 keV to the total number of incident neutrons. The thickness of the coating represented on the “x-axis” and denoted as “ $x_B$ ” corresponds to the thickness of the Boron-coating on the flat cathode, as well as on the flat tops joining the grooves,  $t_{top}$ . The thickness of the layer on the sidewalls of the grooves is given by Eqs. (2) and (3), for the uniform and non-uniform coating scenarios, respectively. Calculations indicate that for a layer thickness below 1.5  $\mu\text{m}$ , the efficiency values are almost identical for both types of cathodes (grooved and flat). The efficiency of the flat cathode reaches a maximum for a layer thickness around 2.5  $\mu\text{m}$ , above which, depending on the irradiation geometry, it either stays constant or it decreases with increasing thickness, in good agreement with the previous calculations [10,20,21]. Since the

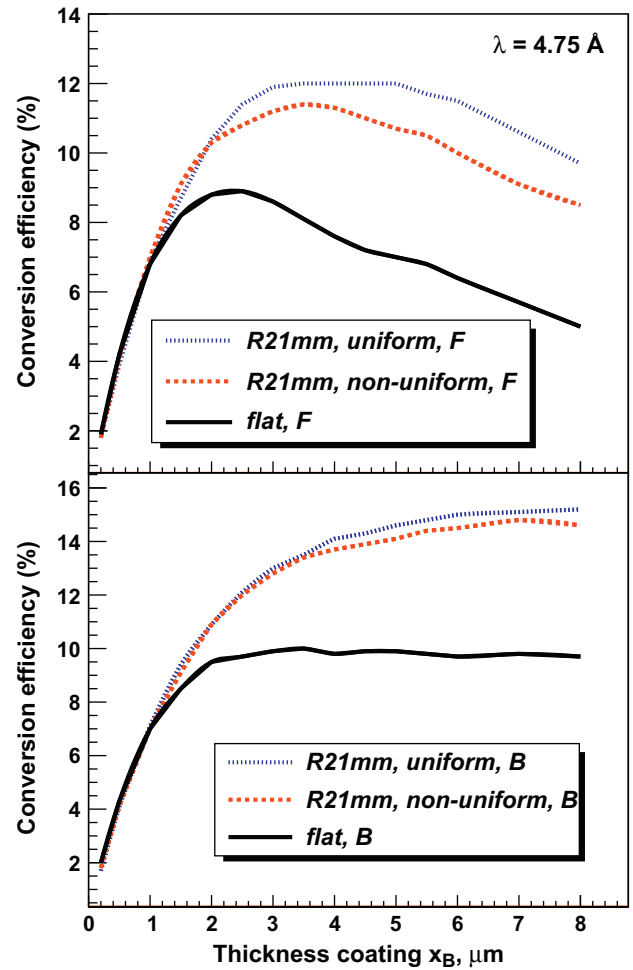


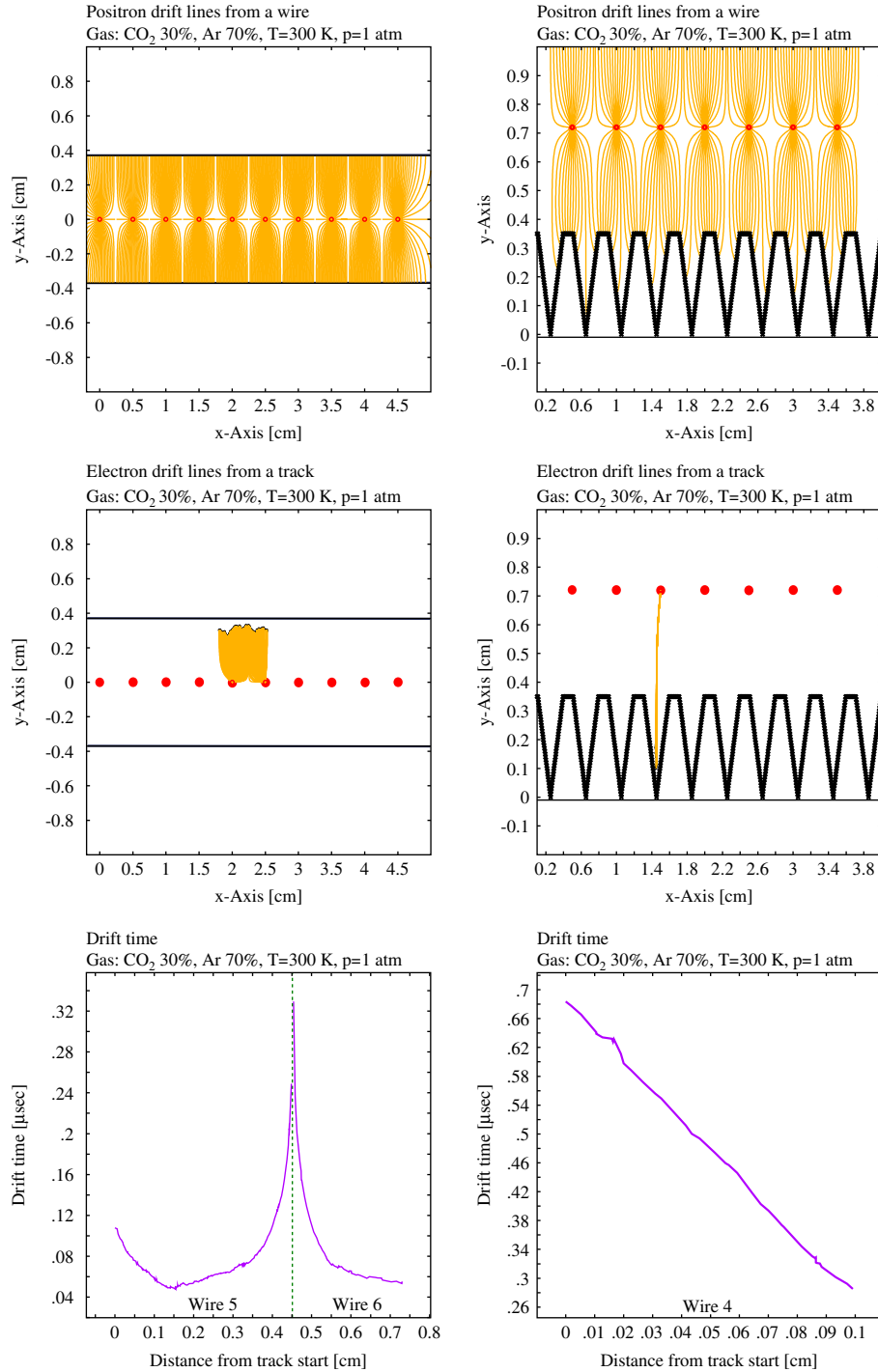
Fig. 6. Simulated conversion efficiencies for the cathodes with 2.1 mm deep grooves and flat surface as a function of the thickness of the coating,  $x_B$ . For the grooved cathode, this parameter represents the thickness of the coating on the flat top, see text for details. The simulations were made for forward- (top panel, marked with “F” in the legend) and backward-irradiation geometries (bottom panel, marked with “B” in the legend). The efficiencies were extracted from the calculated energy deposition spectra with a cut-off at 150 keV.

grooves occupy over 85% of the total surface of the cathode and, according to Eqs. (2) and (3),  $t_{sidewalls} \leq t_{top}$  always, the efficiency calculated for the grooved cathode reaches a maximum for values of the layer thickness that are slightly larger than the optimum value for a single flat cathode. When the thickness of the coating is close to the value that ensures the maximum efficiency for a specific design, the gain in performance resulting from the grooving of the surface of the cathode can be up to 50%.

## 2.2. Electron drift time calculations with GARFIELD

In a traditional MWPC, the anode wires placed halfway between the two parallel cathode plates with flat surfaces ensure a homogeneous electric field near the cathodes. If the flat cathodes are replaced by the grooved cathodes proposed in the present work, the electric field is expected to be disturbed by the millimeter deep macrostructures. This can cause a non-uniform electric field near the surface of the cathode that will influence the drift time of the electrons and, more importantly, affect the charge collection efficiency.

We employed the GARFIELD code version 7.41 [22] to calculate the electric drift lines from the wires and drift times of the low-energy electrons produced by ionizing tracks in proportional



**Fig. 7.** Drift lines and drift times calculated with the GARFIELD code for a MWPC with flat (left) and macrostructured cathodes (right).

counters with flat and macrostructured cathodes. The results of the calculations for the  $\text{ArCO}_2$  (70–30) gas at normal conditions are shown in Fig. 7. The plots on the left column were made for a MPWC with flat cathodes located at  $y = \pm 3.7$  mm. The signal wires, represented in the middle left panel of Fig. 7 by red open circles, were at potential +1.2 kV. The wire pitch was 5 mm. The black wavy line represents the track of an alpha particle with an energy of 1.47 MeV traveling through the counting gas. The electron clusters generated along the ionizing track were calculated by using the energy loss and range tables computed by the SRIM code [28], which were used as input data for the GARFIELD code. The electron transport parameters in the  $\text{ArCO}_2$  gas were

calculated with the MAGBOLTZ code [23] to which GARFIELD is interfaced. The orange lines correspond to the drift lines of the electrons generated along the track. The calculation indicates drift times as large as 340 ns for the electrons generated in the region with low field between two adjacent wires, see bottom left panel of Fig. 7.

In the calculations shown in the right column, the flat cathode planes were replaced by macrostructured cathodes with 3.5 mm deep grooves. For clarity, only the bottom half of the theoretical detector cell is shown in the figure. The orientation of the wires was chosen here to be parallel to the grooves, but calculations indicate similar results for configurations with the wires

perpendicular to the grooves. The starting point of the alpha track was chosen close to the bottom of the groove. According to the calculation, the ionization electrons will be collected by the anode wires ~700 ns after production, see bottom right panel of Fig. 7. This is an increase by roughly a factor two compared to the MWPC with flat cathodes. However, commercial shapers are capable to cope with drift times as large as 2 μs. Therefore, the calculated delay is expected to have little impact on the collection efficiency of the counter.

### 3. Manufacturing of the grooved cathodes

Cathodes with grooves of various heights were fabricated by milling the Aluminium substrate material with a 45° commercial cutter [25]. The substrate material used to fabricate the cathodes consisted of plates of 1.5–6 mm thickness made of the Al–Mg–3 alloy (Al-5754). All plates had a size of 13 × 13 cm<sup>2</sup> in order for them to fit into the small test detector, especially built to test the performance of the various designs. The grooves, however, were milled only on the ~10 × 10 cm<sup>2</sup> central part of the plate. The outer edges were left intact for more stiffness and in order to provide space for the holes used to mount the cathodes inside a small gas detector.

The groove sizes of the cathodes manufactured for proof-of-principle measurements are given in Table 1. For practical reasons, in all investigated designs the ratio between the height of the groove *h* and the length of the flat top *f* was kept near 3. The pitch size given in the fifth column of the table denotes the distance between the centers of two adjacent flat tops and is given by  $f + 2 \cdot h \cdot \tan(\alpha/2)$ . As the fabrication of the grooves by milling became increasingly difficult for *h* < 2 mm, the grooves with *h* = 1.1 mm were milled only on one side of the substrate. 13 × 13 cm<sup>2</sup> flat cathodes were also made of Al–Mg–3 alloy with a thickness of 0.5 mm.

### 4. The coating with boron

The quality of the coating is crucial in order for the Boron-based neutron detectors to be considered a viable alternative for the replacement of the <sup>3</sup>He tubes. Common techniques like electron-beam evaporation or RF/DC magnetron sputtering have an excellent record in providing high quality Boron coatings for large-area neutron detectors [5,24].

As the conformality of the Boron coating is one of the key parameters for the success of the design proposed, an important part of the experimental work was dedicated to the investigation of the capability of an economically feasible coating method like

**Table 1**  
The manufactured cathode designs with 45° grooves. All Aluminium plates have a size of 13 × 13 cm<sup>2</sup>. See Fig. 1 for the definition of the shape parameters *f*, *h*, and *l* given in the columns 2–4. The fifth column gives the distance between the centers of two adjacent flat tops (pitch). Last column gives the number of grooves of specific size per 10 cm.

| Label | Length flat top, <i>f</i> (mm) | Height groove, <i>h</i> (mm) | Thickness Al top, <i>l</i> (mm) | Pitch size (mm) | Grooves per 10 cm |
|-------|--------------------------------|------------------------------|---------------------------------|-----------------|-------------------|
| R5mm  | 1.75                           | 5                            | 1.0                             | 5.9             | 18                |
| R45mm | 1.27                           | 4.5                          | 0.5                             | 5               | 20                |
| R35mm | 1.1                            | 3.5                          | 0.5                             | 4               | 25                |
| R21mm | 0.76                           | 2.1                          | 0.4                             | 2.5             | 40                |
| R11mm | 0.4                            | 1.1                          | 0.4                             | 1.3             | 78                |
| Flat  | –                              | –                            | 0.5                             | –               | 0                 |

magnetron sputtering to deposit a thin, conformal, and contiguous layer of <sup>10</sup>B<sub>4</sub>C (or <sup>10</sup>B) onto the surface of the groove cathodes.

All cathodes listed in Table 1 were coated with enriched <sup>10</sup>B<sub>4</sub>C by DC magnetron sputtering, using a CemeCon 800/9 industrial deposition system [24]. The same system provided the coatings for the large-area Boron detector that is being built by ILL Grenoble and ESS AB in the framework of the CRISP collaboration [7,26]. The deposition parameters used to prepare the grooved cathodes were the same as those used for the coating of the flat Aluminium-blades incorporated the CRISP demonstrator. The description of the coating procedure, film preparation and results of the quality studies are presented in Ref. [24]. Small Silicon samples with smooth surfaces were attached to the grooved cathodes and coated simultaneously. The analysis of the flat Si surfaces provided information about the quality of the Boron coatings such as the adhesion to the substrate, purity, and thickness of the layers. In order to increase the adhesion to the substrate and lower the amount of impurities in the coating, the depositions were made at 400 °C. At this temperature, a density of 2.34 g/cm<sup>3</sup> could be achieved for the Boron coating, which corresponds to 93% of the nominal B<sub>4</sub>C density ( $\rho_0 = 2.52 \text{ g/cm}^3$ ) [24]. Isotope-specific compositional analysis performed with Time-of-Flight Elastic Recoil Detection Analysis (ToF-ERDA) indicated that the coating on the grooved cathodes contains around 98% <sup>10</sup>B, 17 at% C, and 2.4 at% <sup>11</sup>B. Contaminants Mg, Mn, O, H, and N were found to be present in an amount below 1 at%.

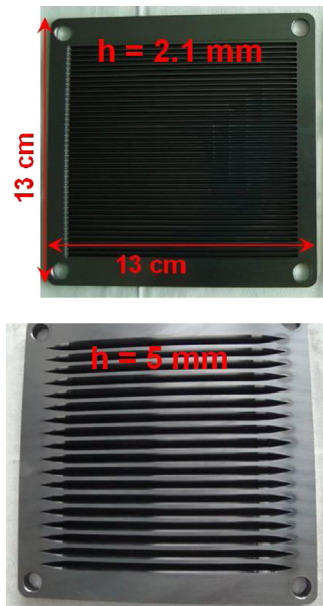
The analysis of the flat silicon-substrates provided a value of 2.95 μm for the thickness of the layers deposited on the surfaces perpendicular to the ion flux, such as the small flat portions joining the grooves. Two flat cathodes were also coated in a separate run with 2.45 μm of the same <sup>10</sup>B<sub>4</sub>C target material.

A second set of flat cathodes was coated with 0.28, 1.2 and 2 μm of <sup>nat</sup>B<sub>4</sub>C by electron-beam evaporation. In order to improve the adhesion of the layer, the surface of the Aluminium-substrate was roughened by sand-blasting before the deposition process.

Table 2 summarizes the target material used for the coatings, <sup>10</sup>B enrichment grade of the target material, thickness of the coating on the surfaces perpendicular to normal incidence, and

**Table 2**  
Summary of the Boron-coated cathodes investigated in the present work. The thickness of the Boron layer quoted for the grooved cathodes corresponds to the thickness of the layer deposited on the small flat tops, see Table 1 for the explanation of the labels.

| Label   | Height groove <i>h</i> (mm) | Target material                 | Enrichment grade <sup>10</sup> B (at %) | Thickness B-layer (μm) | Coating method            |
|---------|-----------------------------|---------------------------------|-----------------------------------------|------------------------|---------------------------|
| R5mm    | 5                           | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.95                   | DC magnetron sputtering   |
| R45mm   | 4.5                         | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.95                   | DC magnetron sputtering   |
| R35mm   | 3.5                         | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.95                   | DC magnetron sputtering   |
| R21mm   | 2.1                         | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.95                   | DC magnetron sputtering   |
| R11mm   | 1.1                         | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.95                   | DC magnetron sputtering   |
| Flat    | –                           | <sup>10</sup> B <sub>4</sub> C  | > 95                                    | 2.45                   | DC magnetron sputtering   |
| Flat-EB | –                           | <sup>nat</sup> B <sub>4</sub> C | 19.9                                    | 0.28, 1.2, 2           | Electron-beam evaporation |



**Fig. 8.** Photographs of the cathodes with 2.1 mm (top) and 5 mm deep grooves (bottom) coated by magnetron sputtering.

the deposition method employed to prepare the cathodes investigated in the present work.

Photographs of the cathodes with 2.1 and 5 mm deep grooves coated by magnetron sputtering are shown in Fig. 8. The large holes in the corners are for mounting the plates in the test detector. The visual inspection with the microscope of the coated surfaces revealed contiguous layers of Boron-converter with a very good adhesion to the Aluminium-substrate for all prepared samples.

## 5. Experimental investigation of the flat and grooved cathodes

The coated cathodes were investigated with a neutron beam of energy 4.75 Å at the TREFF instrument, in operation at the FRM-II research reactor in Munich [27]. The instrument is being used for the characterization of the neutron optics elements, as well as neutron detector prototypes. In the present work, the quality of the coatings and the performance of the various cathodes were investigated by measuring the absorption of the neutron beam in the Boron layer (absorption measurements), and by mounting the cathodes in a small test detector and comparing the observed count rate to that measured with a calibrated  $^3\text{He}$  tube (efficiency measurements).

### 5.1. Characterization of the Boron cathodes in absorption measurements

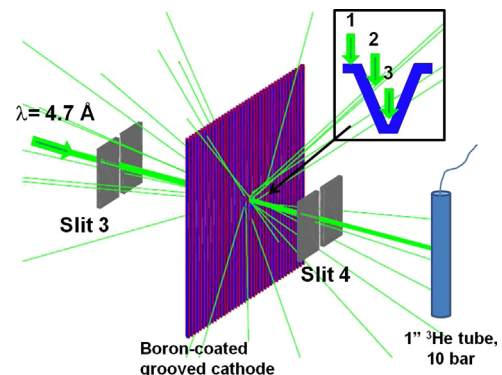
In a neutron absorption measurement, the neutron beam is passed through a sample positioned perpendicular to the incident neutrons. The intensity of the beam is attenuated in elastic and inelastic scattering and neutron capture processes by a factor that depends on the concentration and cross-section of the various materials in the sample. In the case of one Boron layer, the attenuation of the incident neutron beam is mainly due to the absorption in the Boron converter. Although the thickness of the Aluminium substrate material is two orders of magnitude larger than that of the Boron coating (typically 500  $\mu\text{m}$  vs. 3  $\mu\text{m}$ ), the much smaller neutron cross-section of  $^{nat}\text{Al}$  versus  $^{10}\text{B}$  makes the attenuation of the neutron beam in the substrate material

insignificant. Thus, the investigation of the transmission of a collimated, monochromatic beam of incident neutrons through the various cathodes is expected to provide information about the thickness of the Boron coating at the interaction point.

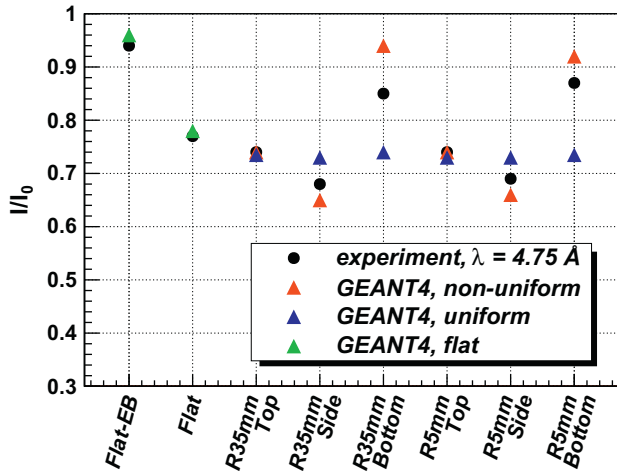
In the present measurements, the focusing and collimation of the incoming neutron beam were accomplished with the help of up to four slits made of thick  $^{nat}\text{B}_4\text{C}$  plates. The layer to be investigated was placed in the front of the fourth slit, perpendicular to the neutron flux, see Fig. 9. The intensity of the incoming neutrons was monitored by means of a fission chamber with an efficiency of  $\sim 10^{-3}$  located in the front of the third slit, while the transmitted beam was measured with an 1 in. tube filled with 10 bar of  $^3\text{He}$  gas and positioned at several centimeters behind the sample. The efficiency of the tube measured for neutrons of 4.7 Å was 98.5%. During the measurements, a Cadmium cover surrounded the tube, providing shielding from the background neutrons. The fractional absorption  $I/I_0$  was extracted from the ratio of the beam  $I$  transmitted through the cathode and the incident neutron flux  $I_0$  measured in the monitor.

Fig. 10 displays the calculated and measured  $I/I_0$  ratios for the Flat-EB cathode coated by electron-beam evaporation with 2  $\mu\text{m}$  of  $^{nat}\text{B}_4\text{C}$  and the Flat, R35mm, and R5mm grooved cathodes coated by magnetron sputtering with 2.45 and 2.9  $\mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ , respectively, see Tables 1 and 2 for the explanation of the labels and details about the coatings. The transmission of the neutron beam through the flat cathodes was measured with a beam spot of  $2 \times 2 \text{ mm}^2$ , which struck the center of the cathode at normal incidence. The absorption of the neutron beam by the two grooved cathodes consisted of transmission measurements through the coatings deposited on the top, on the sidewalls, and at the bottom of the three most central neighboring grooves, see Fig. 9. In order for the incident beam to strike only a specific portion of the coated grooves, the beam spot was reduced to a size that was much smaller than the groove itself. Thus, the investigation of the cathodes with 5 and 3.5 mm deep grooves was made with a beam spot of  $1 \times 2 \text{ mm}^2$  and  $0.5 \times 2 \text{ mm}^2$  (horizontal  $\times$  vertical), respectively. The three data points measured for each of the two investigated designs and marked as “top”, “bottom” and “side” in Figs. 9 and 10 correspond to the average values obtained from each set of several measurements with the beam directed onto the top, bottom and sidewalls of the selected grooves, respectively.

As seen in Fig. 10, the fractional absorption measured for the flat cathodes and tops of the grooved cathodes agrees very well with the theoretical values. As expected, the model makes different predictions for the  $I/I_0$  ratios calculated on the sidewalls and at the bottom of



**Fig. 9.** Schematic representation of the setup for the transmission measurements performed for the grooved cathodes. In the inset, the green arrows marked as 1, 2, and 3 correspond to the positions where the transmission measurements were performed in order to learn about the absorption of the neutron beam on the flat top, side, and bottom of the grooves, respectively. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)



**Fig. 10.** Simulated and measured fractional absorptions  $I/I_0$  for the flat cathode coated by electron-beam evaporation with  $2\text{ }\mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  (Flat-EB) and the flat and grooved cathodes coated by magnetron sputtering with  $2.45\text{ }\mu\text{m}$  and  $2.9\text{ }\mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ , respectively. The absorption of the neutron beam in the Boron layer deposited on the surface of the R35mm and R5mm cathodes was measured on top, on the sidewalls, and the bottom of the three most central neighboring grooves. The experimental data points are given with the statistical errors, which are smaller than the symbol size. In the GEANT4 calculation, the geometry and parameters describing the beam were chosen to closely approximate the experimental situation.

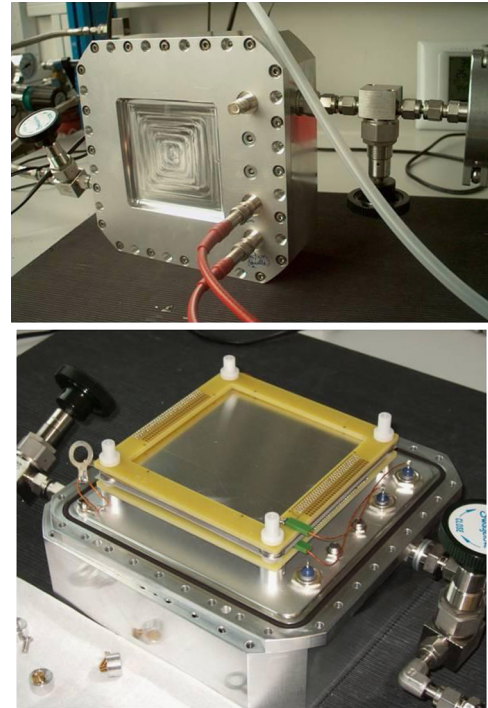
the grooves, depending on the scenario chosen for the coating of the interior surface of the grooves. In the case of the uniform coating, the Boron material is distributed such that the path traveled by the neutron in the converter is the same anywhere on the surface of the groove. This yields the same value for the fractional absorption calculated for the flat tops, sidewalls, and the bottom of the grooves, see blue triangles in Fig. 10. In the non-uniform coating scenario, however, the Boron layer was parametrized such that the thickness of the layer decreases with depth, reaching zero value at the bottom. In this case the thickness of the coating at the mid-distance between the flat top and bottom is slightly larger than that achieved in the uniform coating scenario ( $0.38 \cdot t_{\text{top}}$  vs.  $0.5 \cdot t_{\text{top}}$ , see Eqs. (2) and (3), respectively), which explains the difference between the calculated intensity ratios plotted in Fig. 10 for the sidewalls. At the bottom of the groove the situation is reversed. Here the model predicts a negligible attenuation of the intensity of the beam that strikes a groove coated non-uniformly.

The data presented in Fig. 10 indicates that the fractional absorption measured on the flat tops of the grooves agrees very well with the calculation. The measurements on the sidewalls and at the bottom delivered results that fall at mid-distance between the predictions of the models assuming an ideal, uniform coating of the inside of the groove and the scenario according to which the layer thickness decreases with depth. The comparison with the model calculations suggests that the conformality achieved experimentally by employing DC magnetron sputtering is intermediate between the two coating scenarios depicted in Fig. 3, with the thickness of the layer decreasing near the bottom, but less abruptly than given by Eq. (3).

## 5.2. Characterization of the Boron-lined cathodes in efficiency measurements

### 5.2.1. Investigation of the efficiency of the flat cathodes

For the measurement of the pulse-height spectra and efficiency of the Boron layers, the small, planar detector shown in Fig. 11 was built. The detector has the form of a square parallelepiped with dimensions of  $19.8\text{ cm} \times 19.8\text{ cm} \times 6.8\text{ cm}$ . It can accommodate up to six  $13 \times 13\text{ cm}^2$  flat cathodes, which can be mounted to create three independent MWPCs. The neutron beam struck the cathodes



**Fig. 11.** Photographs of the test detector built in the present work to assess the performance of the various cathode designs. The photograph in the top panel shows the detector closed and ready to take data. Visible in the picture are the thin entrance window, the SHV feedthroughs and the inlet lines, and valves for the detector evacuation and gas fill purposes. The photograph in the bottom panel displays the inside of the detector, where one of the small anode wire planes used to amplify and collect the signal is visible.

of the MWPCs through a  $10 \times 10\text{ cm}^2$  thin entrance window. The signal generated at the interaction of the neutrons with the Boron coatings was amplified and collected by means of anode wire planes located symmetrically at a distance of  $3.7\text{ mm}$  between each two cathodes. The anodes were made of  $20\text{ }\mu\text{m}$  diameter tungsten wires soldered on printed circuits. For the tests presented in this paper, two types of anode planes were fabricated. The anode planes with the wires separated by  $2\text{ mm}$  provided a very uniform electric field leading to a constant gas gain over the whole cathode surface and pulse-height spectra with a very good energy resolution, which makes it easy for the comparison with the model calculations. The anode planes with the wires separated by  $5\text{ mm}$  provided a less uniform field near the cathodes and hence, to a detector response with poorer energy resolution. However, such a pitch size is more realistic for an application whose ultimate goal is the realization of an affordable large-area neutron detector.

The anodes and cathodes were stacked together on the detector lid by using long plastic screws and spacers, which allowed for an easy mounting of the cathodes to be investigated. The detector was operated in a continuous flow of  $\text{Ar}/\text{CO}_2$  gas (70/30). The sensitive area of each MWPCs was  $7 \times 10\text{ cm}^2$ , which corresponds to the surface area covered by the anode wires. The detector response to radiation was tested with the  $^{55}\text{Fe}$  and  $^{241}\text{Am}$  sources placed inside the detector. No position information can be obtained with the current detector setup.

The cathode designs to be studied were mounted one by one in the MPWC closest to the neutron source. The measurements presented in this paper were performed with the cathodes held at ground potential and the anode wires at positive bias voltages applied through a SILENA 205 charge-sensitive amplifier. Typical working voltages were  $\sim 1.5\text{ kV}$  when the cathodes had flat surfaces and  $\sim 1.6\text{--}1.7\text{ kV}$  when measured with the grooved cathodes. The output signal of the anode was first shaped by a

SILENA 7614 amplifier with a shaping time of 1  $\mu$ s and then fed to a multichannel analyzer, which had a dynamic range of 5 V.

The layers coated with 0.28, and 1.2  $\mu$ m of  $^{nat}\text{B}_4\text{C}$  and 2.45  $\mu$ m of  $^{10}\text{B}_4\text{C}$  deposited onto 0.5 mm flat Aluminium substrates (labeled as *Flat-EB* and *Flat* in Table 2) were investigated with the 4.7  $\text{\AA}$  parallel beam, and the measured corresponding pulse-height spectra and absolute efficiencies were used to test the predictive accuracy of the GEANT4 calculations. These model validation measurements with the flat cathodes are important because they establish the capability to predict the performance of cathodes with more complex geometry like those proposed in the present work. During the measurements, the intensity of the incident neutron beam was monitored by means of a fission chamber mounted upstream of the test detector. The spectra were measured with the anode plane with 2 mm wire pitch. The bias voltage was adjusted in order to have the endpoint of the pulse-height spectra around channel 1800. The acquisition time for the pulse-height spectrum taken with the 0.28  $\mu$ m sample was 1000 s, while the spectra of the 1.2, 2, and 2.45  $\mu$ m thick coatings were collected for 600 s each.

Fig. 12 shows the comparison between the experimental pulse-height spectra collected with the anode plane with the 2 mm wire pitch size and calculated energy deposition spectra for layers coated with 0.28 and 1.2  $\mu$ m of  $^{nat}\text{B}_4\text{C}$ , and 2.45  $\mu$ m coating of enriched  $^{10}\text{B}_4\text{C}$ . For illustration purposes, the number of incident neutrons in the calculations was chosen such that the distributions of the observed and simulated events are similar. For the spectra displayed in the top panel of the figure, the thickness of the Boron layer is much smaller than the ranges of the reaction products in Boron, therefore the energy lost in the layer is not significant. This makes the full-energy peaks at 840 and 1470 keV, corresponding to the  $^7\text{Li}$  ion and the alpha particle, respectively, from the most intense branch of the  $n+^{10}\text{B}$  reaction, clearly visible in the spectra. The calculated spectrum was convoluted with a Gaussian distribution of width 0.034 keV, in order for the FWHM of the peak centered at 1470 keV to agree with the experimental observation. The long tail at energies below 150 keV in the experimental data arises from the  $\gamma$ -background. The neutron- and  $\gamma$ -events are very well separated, therefore the loss in efficiency due to the  $n$ - $\gamma$  threshold is negligible.

Increasing the thickness of the coating, the charged particles released in the reactions that take place more than several hundreds of nanometers away from the gas interface will have to travel large distances in order to escape the layer, thereby losing an important amount of their kinetic energy. This leads to a substantial smearing of the individual peaks associated with the two reaction products. In this case the pulse-height spectra show a step-like function continuing toward low energies, see the middle and bottom panels of Fig. 12. Good agreement between the measurement and the calculation is observed for both the investigated samples. The model thus reproduces well the spectral features and tendency observed experimentally.

Fig. 13 displays the comparison between the calculated and the measured efficiencies for all four samples. The experimental efficiencies were obtained by normalizing the integrals of the measured pulse-height spectra to the response of a small  $^3\text{He}$  tube of known efficiency, obtained by removing the test detector from its position and placing the tube instead. As seen in Fig. 13, the calculation fits very well the experimental data. As the simulation does not include the electric field effects, the good agreement between the measured and the calculated efficiencies suggests that the charge collection efficiency is close to 100%.

### 5.2.2. Investigation of the conformity of the Boron coating on the surface of the grooved cathodes with a collimated neutron beam

The conformity of the boron coatings on the surface of the grooved cathodes was investigated by mounting the cathodes in the test detector and measuring the variation in efficiency across

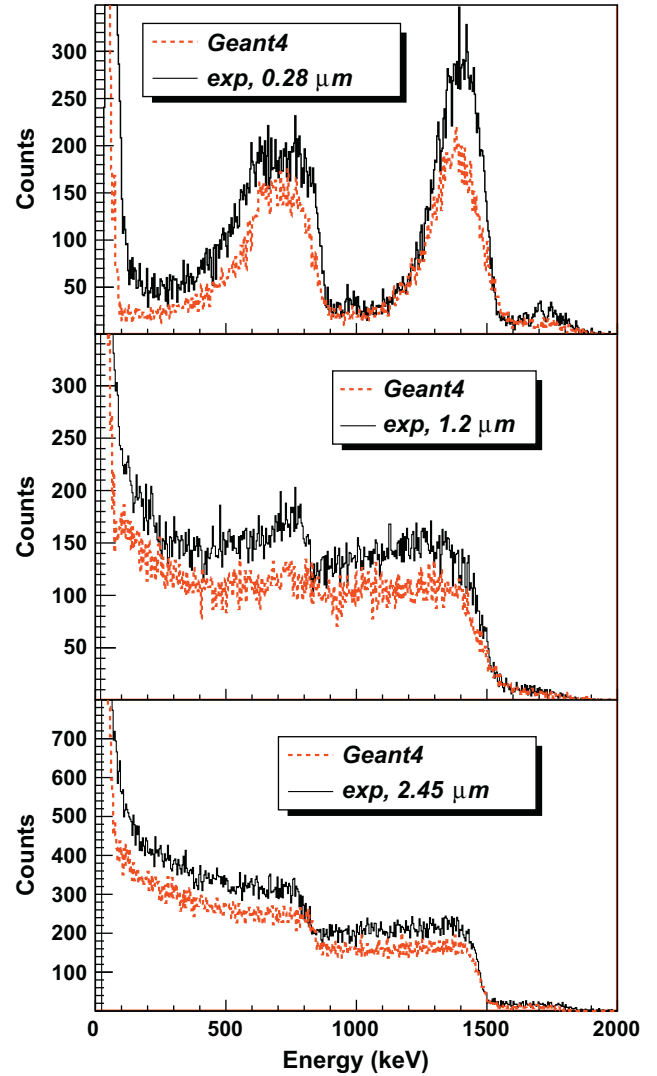


Fig. 12. Comparison between simulated energy deposition spectra (red dashed line) and measured pulse-height spectra (full black line) for the 0.28  $\mu$ m (top) and 1.2  $\mu$ m (middle) coatings of  $^{nat}\text{B}_4\text{C}$ , and 2.45  $\mu$ m (bottom) coating of enriched  $^{10}\text{B}_4\text{C}$  deposited on flat aluminum-substrates. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

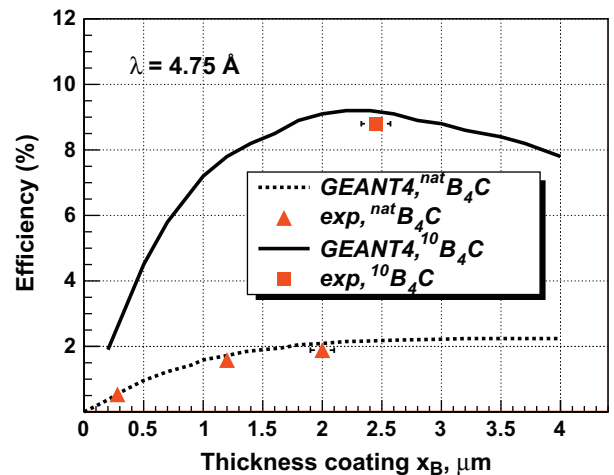


Fig. 13. Simulated and measured efficiencies for the flat cathodes coated by electron-beam evaporation with 0.28, 1.2, and 2  $\mu$ m of  $^{nat}\text{B}_4\text{C}$  and by magnetron sputtering with 2.45  $\mu$ m of enriched  $^{10}\text{B}_4\text{C}$ .

the surface of the grooves. In order to perform such an efficiency scan accurately, the beam spot had to be reduced to a size that is much smaller than the size of the groove to be investigated. The collimating slit located immediately after the monochromator and the one in the front of the detector were set to a size of  $5 \times 2.5$  mm (horizontal  $\times$  vertical) and  $0.2 \times 2$  mm, respectively, such that a horizontal beam divergence of 0.1 mm was achieved for a distance between the two slits of 3.59 m.

The test detector was placed on a X–Y moving table at a distance of 6 cm between the surface of the entrance window and the last collimating slit. The grooved cathode to be investigated was mounted inside the detector with the grooves extending along the Y-axis. Similar to the absorption and efficiency measurements reported above, the intensity of the incoming beam was monitored by means of a fission chamber mounted half way between the monochromator and the detector. The experimental efficiencies were determined by normalizing the number of events observed in the pulse-height spectra to the integral of the spectrum recorded with a 1 in. tube filled with 10 bar of  $^3\text{He}$  gas mounted in the detector position.

With this setup, the conformity of the Boron coating was investigated on the surfaces of the R5mm, R35mm, and R21mm cathodes. Before starting the in-beam measurements, the moving table was calibrated such that the  $x=0$  and  $y=0$  positions were at the center of the entrance window of the detector. This (0,0) reference point corresponds to an arbitrary position on the surface of the groove. Measurements were performed by moving the scanning table along the X-axis in steps of 0.5 mm for the R5mm and R35mm cathodes and 0.25 mm for the R21mm cathode, and pulse-height spectra were acquired at each step. The measurements were performed with the anode plane with the 5 mm wire pitch size mounted 3.7 mm away from the flat tops of the grooves. All measurements were made with the anode wires perpendicular to the grooves. However, earlier tests showed that the position of the wires relative to the grooves (either parallel or perpendicular) does not play any role for the efficiency of the investigated designs.

Fig. 14 displays examples of the detector response when the collimated neutron beam struck the middle of the top, the sidewall at two different distances from the top, and the bottom of one of the central grooves of the R5mm (top panel) and R21mm cathodes (bottom panel). Also shown are the spectra measured by placing the detector incorporating the cathodes to be studied in the gamma field produced by a moderated Cf source, which is a copious  $\gamma$ -emitter. The  $\gamma$  dose-rate from the source was  $\sim 30$  mR/h. As it can be seen in both panels, the scanning of the inside of the grooves yielded pulse-height spectra with very distinct spectral features when compared to the structureless spectra collected on the flat top. The peak-like structures observed around channels 200 and 300 in the energy spectra measured at two different distances on the sidewall originate from the reaction products that lose only part of their energy in the counting gas before implanting into the opposite wall of the groove. This amount of energy depends on the position of the interaction point of the incident neutron with the Boron layer on the sidewall. For interactions with the coating at approximately mid-distance between the top and bottom of the groove, the distance to the opposite wall is around 2 mm for the R5mm cathode, and  $\sim 0.8$  mm for the R21mm cathode. According to the SRIM calculations, alpha and  $^7\text{Li}$  ions with average kinetic energies around 1 and 0.5 MeV, respectively, lose  $\sim 0.35$  MeV per millimeter when transiting an Ar/CO<sub>2</sub> gas (70/30) at room temperature [28]. This suggests that the structures occurring in the pulse-height spectra shown in red in Fig. 14 correspond to energies around 700 and 300 keV for the R5mm and R21mm cathodes, respectively.

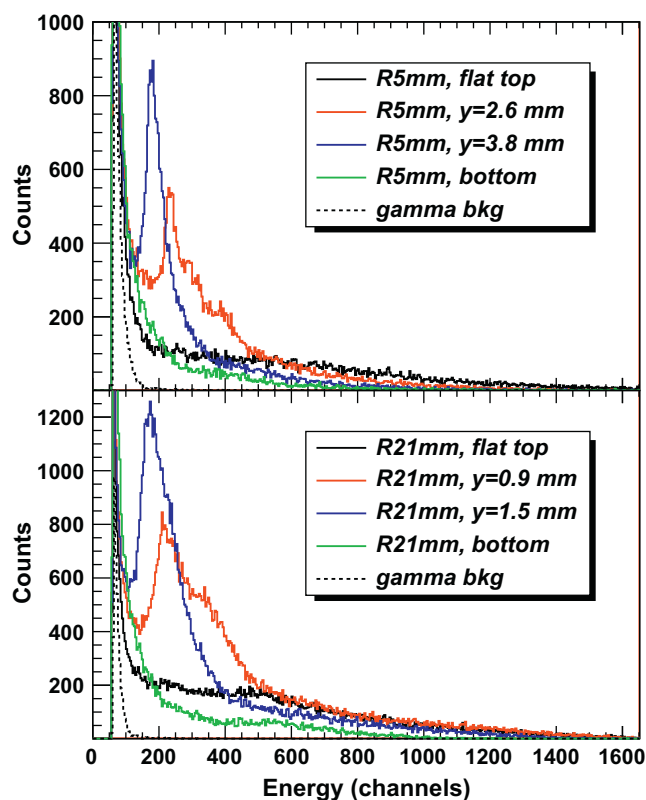
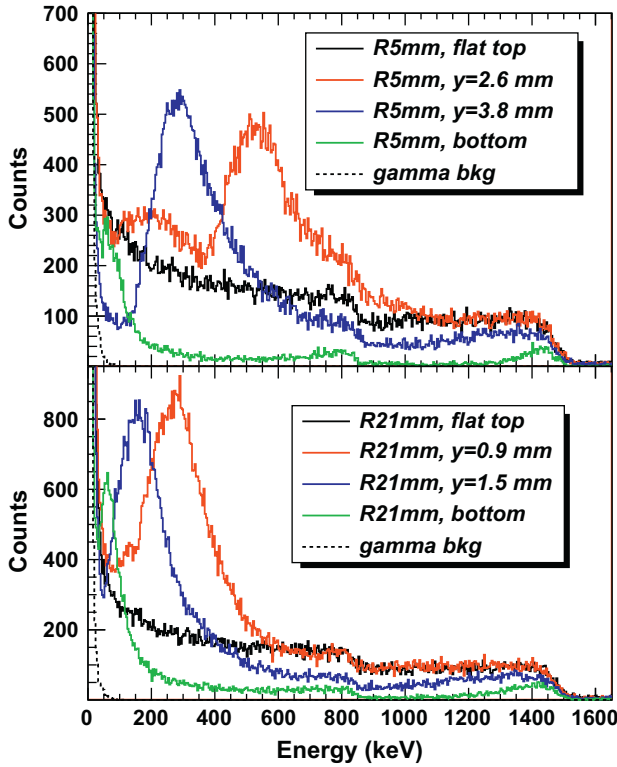


Fig. 14. Top: experimental pulse-height spectra obtained with the neutron beam striking the top, the sidewall at  $y=2.6$  mm and 3.8 mm vertical distances from the top, and the bottom of one of the central grooves of the R5mm cathode. Bottom: same as above, but for the R21mm cathode. The  $\gamma$ -background spectra accumulated by exposing the detector to the  $\gamma$ -field emitted by a moderated  $^{252}\text{Cf}$  source are plotted with dashed black lines. The signal was collected with the anode plane with 5 mm wire pitch.

In Fig. 15 the experimental pulse-height spectra are compared to the results of the GEANT4 simulations. Calculations were made for both coating scenarios illustrated in Fig. 3, but only those obtained by assuming that the thickness of the coating deposited on the sidewalls of the grooves decreases with decreasing the distance from the bottom are shown in Fig. 15. The number of simulated neutrons was chosen such that the statistics in the calculated energy spectra was similar to that collected experimentally. No folding of the calculated energies with a Gaussian function was performed in order to broaden the distributions. As it can be seen in Figs. 14 and 15, the low energy part of the signal collected from the sidewalls, exhibiting the partial-energy peak, is well reproduced by the calculation. The comparison with the model calculations suggests that the peaks observed around channels 200 and 400 in the experimental spectra correspond to the energies of 300 and 600 keV, respectively, see the red and blue spectra in the top panel of Fig. 14. In the case of the R21mm cathode, the comparison between the calculated and experimental spectra measured inside the groove suggests that the partial-energy peaks occurring at channels 200 and 300 correspond to energies of 150 and 300 keV, respectively, see bottom panels of Figs. 14 and 15.

The energy deposition spectra calculated with the beam striking the bottom of the two cathodes and represented with green line in Fig. 15 have poor statistics and consist of three major structures: the peak around 100 keV arising from the wall-effect discussed above and the two steps corresponding to the 840-keV Lithium and 1470-keV alpha ions from the main  $n+^{10}\text{B}$  reaction branch. In the measured spectra, the low-energy peak is overlapped with the much stronger signal from the  $\gamma$ -background and hence, indistinguishable.

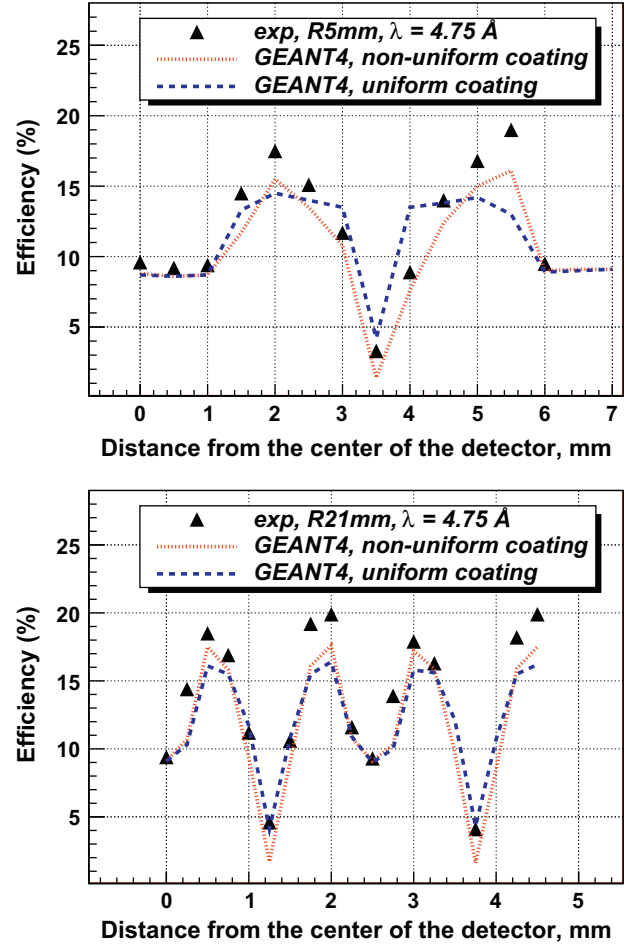


**Fig. 15.** Top: simulated energy deposition spectra obtained with the neutron beam striking the top, sidewall, and the bottom of one of the central grooves of the *R5mm* cathode. Bottom: same as above, but for the *R21mm* cathode. In the simulation the beam parameters, counting gas, and setup were chosen to closely resemble the experimental conditions.

In this critical region, the wall-effect is substantial. According to the calculation, over 60% of the neutron-related events have energies below the discriminator threshold set at 150 keV. As a result, the events that can generate a valid neutron signal and contribute to the overall efficiency are mainly those emitted almost perpendicular to the groove and moving towards the anode wires.

Fig. 16 shows the efficiencies extracted from the pulse-height spectra obtained from the scans with the collimated neutron beam. The efficiencies were determined by considering only the events above channel 150, which the comparison of the measured and simulated energy spectra suggests that it corresponds to an energy  $\sim 150$  keV. As it can be seen in the figure, the measured values vary between 3% and 20%. The current position of the beam on the surface of the groove was easily identified during the scans from the appearance of the pulse-height spectra, which, as seen in the examples shown in Fig. 14, exhibit very distinct spectral characteristics. The structureless, high-gain spectra were immediately associated with the flat top of the grooves and those showing well-formed peaks at medium energies were linked to the sidewalls. A reduced count rate was the indication that the neutron beam was striking the bottom of the groove. The conversion of the integrals of the observed spectra into efficiencies revealed values around 9% for the flat tops, between 9 and 20% for the sidewalls, and  $\sim 3\%$  for the bottom, see Fig. 16. The efficiencies measured for the flat tops of both investigated designs agree very well with the value measured for the flat cathode coated with  $2.45 \mu\text{m}$  of the same material and the predictions of the GEANT4 simulations shown in Fig. 13.

Also included in Fig. 16 are the results of two sets of GEANT4 calculations, one for each coating scenario depicted in Fig. 3a and b. The comparison between the calculation and the measurement suggests that the simulation assuming a non-uniform coating of the groove fits better the data obtained for the *R5mm* cathode. For



**Fig. 16.** Top: simulated and measured efficiencies obtained by scanning the surface of the cathode with 5 mm deep grooves (*R5mm*) coated on the flat top with  $2.95 \mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$  by magnetron sputtering. The experimental efficiencies were extracted by normalizing the number of counts observed in the pulse-height spectra acquired at each position to the number of events in the spectrum of a calibrated  $^3\text{He}$  tube mounted in the detector position. Bottom: same as above, but for the *R21mm* cathode.

the *R21mm* cathode, the difference between the predictions of the two coating scenarios becomes less pronounced. In this case, the smaller size of the groove leads to a steeper decrease of the thickness of the coating with depth when compared to the cathode with larger grooves. The good agreement between the results of the efficiency measurements and GEANT4 simulations assuming 100% efficiency for charge collection suggests that the weaker electric field inside the groove does not affect the collection of the charge significantly.

The measurements reported in this subsection showed clear evidence that the sidewalls of the grooves are zones with enhanced efficiency, confirming the theoretical calculations, which predict larger neutron sensitivities for converter layers deposited on inclined surfaces, owing to the higher probability for the reaction products to escape the layer. The experimental data also indicate that the low efficiencies measured at the bottom of the groove are due to a significant wall-effect and a poor coverage with Boron, both aspects being well accounted for by the calculation.

### 5.2.3. Investigation of the overall efficiency of the grooved cathodes with a monoenergetic neutron beam

The overall performance of the proposed designs was investigated with the same setup used to perform the efficiency scans.

The evidence presented and discussed in the previous subsection showed that the grooved cathodes consist of zones with variable efficiencies, depending on the position on the surface of the groove where the interaction takes place. Therefore, for an accurate determination of the overall performance of the cathode, the size of the beam spot has to be at least as large as the groove pitch. The size of the beam spot was adjusted by opening the collimator in the front of the detector to  $3.7 \times 2$  mm (horizontal  $\times$  vertical), which led to a beam spot size in the horizontal direction of 5.2 mm when the beam divergence is taken into account. With this beam spot striking the investigated cathode, several measurements were performed by moving the table horizontally in steps of 1 mm. Energy deposition spectra were recorded at each step and the efficiency was determined from the sum of the observed events normalized to the counts measured in the same experimental conditions by a 1 in.  $^3\text{He}$  tube of known efficiency mounted in the detector position. The overall efficiency of the cathode was calculated as the average of the efficiency values measured at each step.

Fig. 17 presents the pulse-height spectra collected for equal times and in the same experimental conditions with the *R5mm* (full black line), *R21mm* (dashed red line) and *Flat* cathodes (dashed blue line). The comparison of the three experimental pulse-height distributions indicates that the spectra measured with the grooved cathodes contain extra counts below 1 MeV. These events arise from the reaction products with restricted line-of-sight generated in the reaction of the incident neutrons with the Boron converter inside the groove, and are responsible for the enhancement of the overall efficiency with respect to the flat cathode. The slope of the distribution of the sidewall events depends on the height of the groove, which determines the distance available in the counting gas that the charged ions generated on one side of the groove can travel before implanting in the opposite wall. For the cathode with large grooves, this distance measures up to several millimeters, which is large enough for the ions to deposit a large part of their kinetic energy and give rise to signals as high as 1 MeV. It is worth mentioning that the shape of the spectrum recorded with the *R21mm* grooved cathode is very similar to spectrum obtained with the coated star-shaped straws and presented in Fig. 9(a) of [16].

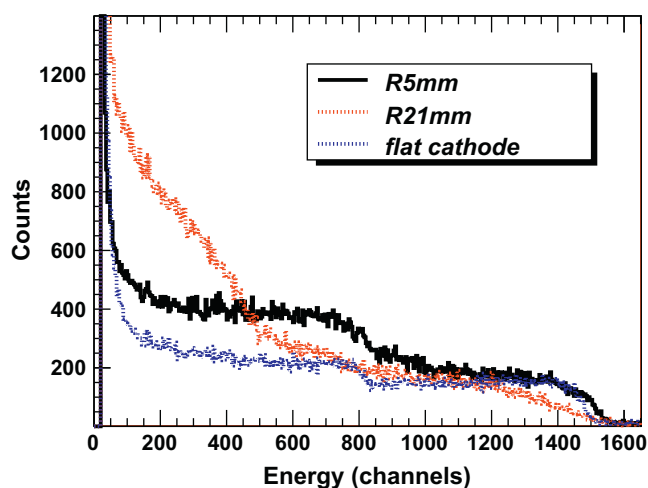


Fig. 17. Experimental pulse-height spectra for the *R5mm* (full black line), *R21mm* (dashed red line) and a flat cathode (dashed blue line) coated with enriched  $^{10}\text{B}_4\text{C}$  by magnetron sputtering. The thickness of the coating on the flat tops of the grooved cathodes was  $2.95 \mu\text{m}$ , that of the layer deposited on the surface of the flat cathode was  $2.45 \mu\text{m}$ . All cathodes were measured for equal times with a  $\lambda=4.7 \text{ \AA}$  neutron beam at normal incidence. The signal was collected with the anode plane with 2 mm wire pitch. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

The results of the GEANT4 simulations corresponding to the measurements shown in Fig. 17 are presented in Fig. 18. In order to facilitate the comparison with the experiment, the spectrum calculated for the flat cathode was drawn in each panel of Fig. 18. The model describes well the spectral features observed in all three experimental pulse-height spectra. The calculations assuming uniform coating of the sidewalls yielded to higher statistics at low energies, owing to the larger thickness of the coating near the bottom of the groove.

The overall efficiencies obtained from the average of the efficiency values measured on the surface of the cathode in several points along the “x-axis” are plotted in Fig. 19, along with the estimation of the GEANT4 model assuming uniform (full red line) and non-uniform (dashed green line) coating of the sidewalls. The efficiencies are represented as a function of the height of the groove,  $h$ . All grooved cathodes were coated with  $2.95 \mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ , as measured on the flat tops of the groove. The dashed-dotted blue line corresponds to the efficiency value calculated for a flat cathode of the same active area coated with  $2.45 \mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ , see also Fig. 13. As observed in Fig. 19, the differences between the results obtained for the two coating scenarios are not important for the grooves with small sizes, and both fit well the experimental data points obtained for the cathodes with grooves of heights 1.1, 2.1, and 3.5 mm. The measurements with the cathodes with large groove sizes are better fitted by the calculation assuming a non-uniform coating of the sidewalls. This is in line with the results of the scan data presented in the previous subsection. Besides, although the efficiency scans showed that the electric field effects have only a minor effect on the efficiency values, the comparison of the

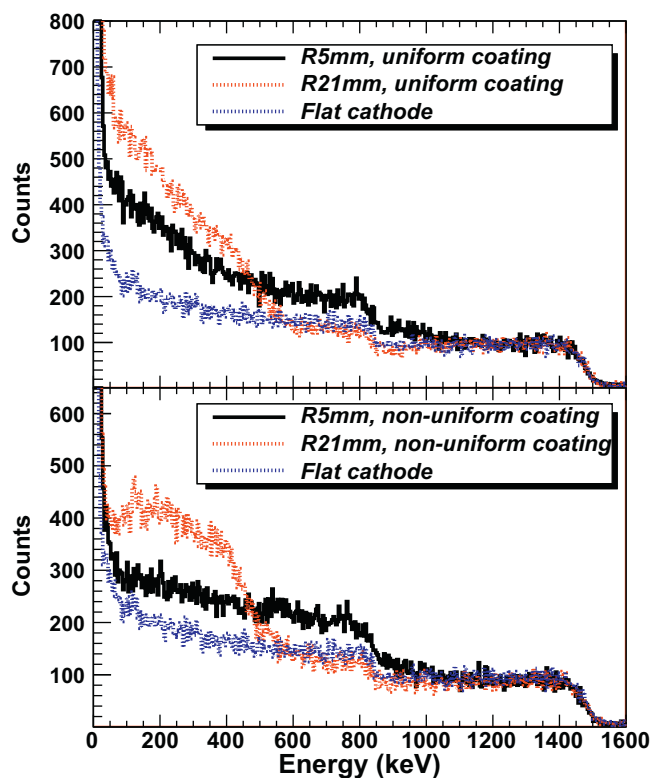
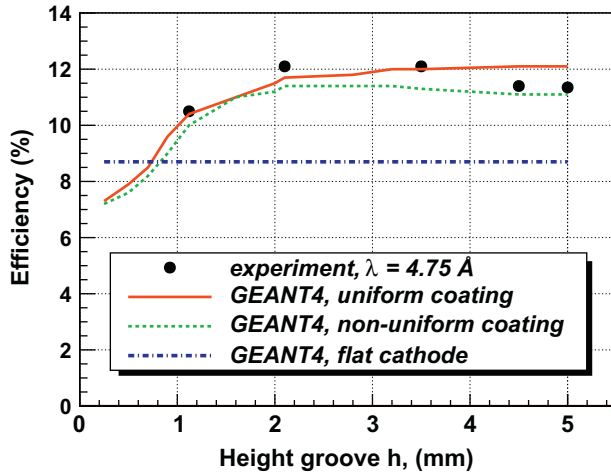


Fig. 18. GEANT4 energy deposition spectra for the *R5mm* (full black line) and *R21mm* (dashed red line) coated with  $2.95 \mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ . The calculation for the grooved cathodes was performed assuming a uniform coating of the sidewalls, see Fig. 3a. For the spectra presented in the bottom panel the thickness of the coating was assumed to decrease with depth as sketched in Fig. 3b (non-uniform coating). The spectrum calculated for a flat cathode coated with  $2.45 \mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$  is drawn in both panels for comparison purposes. The setup used in the simulations resembled closely the experimental conditions.



**Fig. 19.** Simulated and measured efficiencies for the cathodes with grooves of various sizes. The measurements were performed with collimated neutron beam of 4.7 Å orthogonal to the cathode surface. All grooved cathodes investigated experimentally were coated by magnetron sputtering with 2.95  $\mu\text{m}$  of enriched  $^{10}\text{B}_4\text{C}$ , as measured on the flat tops of the grooves. The horizontal line corresponds to the efficiency value calculated for a flat cathode coated with 3  $\mu\text{m}$  enriched  $^{10}\text{B}_4\text{C}$ .

experimental data obtained for the grooves of heights 2.1 and 5 mm indicated that the loss in efficiency caused by the weaker electric field at the bottom is more important for the cathode with large grooves. This suggests that grooves with heights  $h \approx 2$  mm are optimal and should be considered for use in a future detector prototype.

## 6. Calculated efficiency for multilayer configurations

In the previous sections we demonstrated experimentally that the neutron sensitivity of a Boron-layer can be increased by creating grooves with a depth of a few millimeter in the substrate material before the Boron converter is coated on its surface. However, a realistic detector must incorporate several Boron layers in order to represent a viable alternative for the replacement of the  $^3\text{He}$  tubes in large-area detectors for neutron scattering applications. Fig. 20 shows the dependence on the thickness of the Boron converter of the cumulated efficiency of configurations incorporating stacks of 2, 5, 10, and 20 cathodes with 2.1 mm deep grooves (full lines) and flat surfaces (dashed lines). In the calculation, the cathodes were “coated” on both sides with a thin layer of  $^{10}\text{B}_4\text{C}$ . For the macrostructured cathodes, the parameterization given by Eq. (3) was used for the conformality of the coating. Energy deposition spectra were calculated for each stack and the efficiency values plotted in Fig. 20 were extracted by setting the  $n$ - $\gamma$  threshold at 150 keV. For the macrostructured cathodes, the value quoted on the “X”-axis corresponds to the thickness of the Boron coating measured on the flat tops joining the grooves,  $t_{\text{top}}$ . The calculations were performed for neutrons with a wavelength of 1.8 Å, which struck the cathodes at normal incidence.

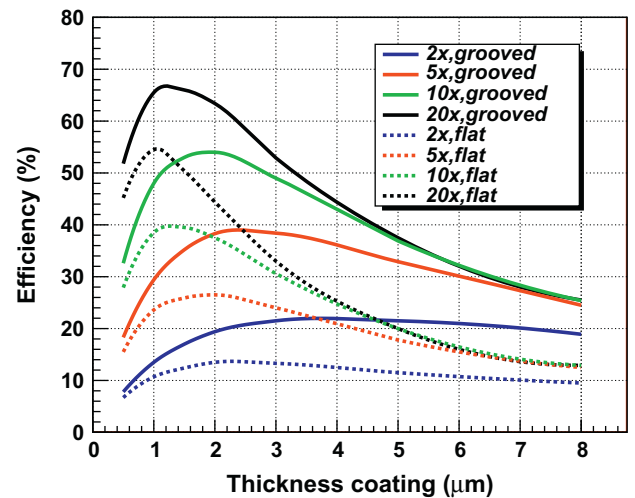
As it can be seen in Fig. 20, each of the calculated efficiency curves reaches a maximum for a specific optimum thickness, which decreases as the number of the layers in the stack increases. The optimum thickness for a stack of macrostructured layers is in general higher than that of a stack incorporating the same number of flat cathodes. The reason for this difference is that for the macrostructured cathodes the optimum value for the thickness of the Boron converter is driven by the thickness of the layer on sidewalls of the grooves, which makes  $\sim 70\%$  of the total active area of the cathode. For the calculations shown in Fig. 20, this

thickness was assumed to be proportional to the ratio  $t_{\text{top}}/ (2.1 \text{ mm})$ , see Eq. (3) and Fig. 3b. The calculation also indicates that the optimum thickness of the Boron converter in multilayer configurations decreases with increasing the wavelength of the incident neutrons. The results obtained for the stacks populated with 5 and 10 flat double-sided coated cathodes agree well with the predicted efficiencies for 1 and 2 modules, each incorporating 5 rows of coated straws [29].

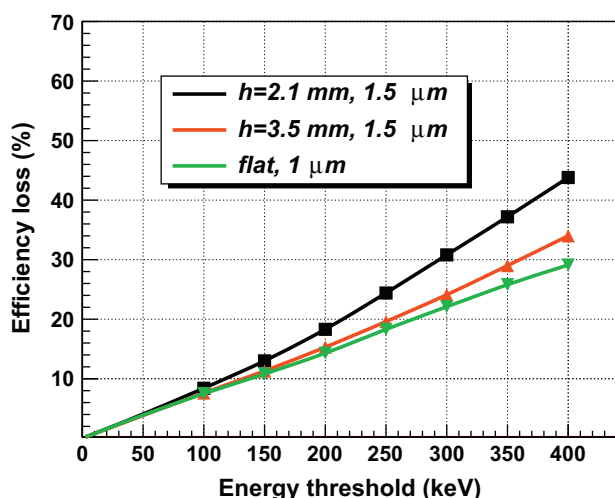
The GEANT4 model predicts that a stack incorporating 10 macrostructured cathodes coated on both sides with  $\sim 1.5 \mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  can achieve an efficiency similar to that of a stack of 20 cathodes with flat surface and coated on both sides with  $\sim 1 \mu\text{m}$  of  $^{10}\text{B}_4\text{C}$ . For a realistic prototype based on the MWPC technology and designed to reach an efficiency of  $\sim 50\%$  for thermal neutrons, the use of macrostructured cathodes instead of flat coated plates leads to a reduction by approximately 50% of the depth of the detector by reducing the number of individual proportional counters and consequently, number of electronics channels.

Fig. 21 shows the efficiency loss factor due to the  $n$ - $\gamma$  threshold for stacks of 10 macrostructured and 20 flat cathodes coated on both sides with  $^{10}\text{B}_4\text{C}$ . Calculations were performed for cathodes with 2.1 and 3.5 mm deep grooves and a thickness of the Boron converter of 1.5  $\mu\text{m}$ . The thickness of the Boron layer considered in the calculation for the stack of 20 cathode planes with flat surface was 1  $\mu\text{m}$ . The calculation indicates that by setting the  $n$ - $\gamma$  threshold at the value of 150 keV, which was determined experimentally to ensure a proper  $\gamma$  rejection for the data presented in this paper, causes a loss in efficiency between  $\sim 10\%$  for the stacks of flat cathodes and  $\sim 13\%$  for the stack of ten cathodes with 2.1 mm deep grooves. When the threshold is raised above 150 keV, the stronger wall effect for the reaction products generated inside the 2.1 mm grooves leads to a larger loss in efficiency compared to the cathodes with larger groove size or flat surface. In practice, threshold values at 100 keV and 30 keV were reported for a detector incorporating 14 double-sided coated blades and a module consisting of  $10 \times 5$  coated straws, respectively [2,7].

As it can be seen in Fig. 21, the deterioration of the efficiency with increasing the threshold level can be slowed down by increasing the size of the groove. The calculation indicates that by setting the  $n$ - $\gamma$  threshold to 300 keV, the efficiency of a stack of



**Fig. 20.** Calculated efficiency for stacks of 2, 5, 10, and 20 cathodes plates with 2.1 mm deep grooves (full lines), and with a flat surface (dashed lines) as a function of the thickness of the  $^{10}\text{B}_4\text{C}$  layer. The Boron converter was applied on both surfaces of the cathodes. The efficiencies were extracted from the calculated energy deposition spectra with an energy cut-off at  $E = 150 \text{ keV}$ . Calculations were performed for neutrons with a wavelength of 1.8 Å, striking the surface of the stack at normal incidence.



**Fig. 21.** Calculated efficiency loss factor due to energy threshold for stacks of 10 macrostructured (grooved) and 20 flat cathodes plates coated on both sides with 1.5  $\mu\text{m}$  and 1  $\mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  layer, respectively.

20 cathode planes coated on both sides with 1  $\mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  is reduced from 55% to 49%, that of a stack of 10 macrostructured cathodes with 3.5 mm deep grooves coated on both sides with 1.5  $\mu\text{m}$  of  $^{10}\text{B}_4\text{C}$  decreases from 53% to 46%, while the efficiency of a stack of 10 cathodes with 2.1 mm deep grooves decreases from 53% to 44%, see Figs. 20 and 21. Thus, the calculation indicates that a large  $\gamma$ -threshold has just a small effect on the overall efficiency of a multilayer detector, for which the use of macrostructured cathodes is still advantageous. The choice of the size of the groove for a future demonstrator based on the proposed concept must be a trade-off among the uniformity of the coating and delayed charged collection inside the groove, both favoring small groove sizes ( $h \sim 2$  mm), and the minimization of the efficiency loss while maximizing the  $\gamma$ -suppression level, which improves with increasing the size of the groove. The final design of the macrostructured cathode will depend on the specific requirements of the applications for which this concept might prove to be a technological option.

## 7. Conclusions

In this paper we presented and discussed a novel design concept for a macrostructured cathode that can be coated with a thin layer of the  $^{10}\text{B}$ -converter and mounted to replace the Boron-lined flat parallel plates in a MWPC used for slow neutron detection. The proposed design consists of a 3D regular pattern with millimeter deep grooves created in the substrate material of the cathode. The GEANT4 simulations predicted that the patterning, followed by the coating with Boron of the surface of the cathode could bring an increase in the neutron detection efficiency of up to 40% with respect to a flat surface of the same area coated with a Boron layer of roughly the same thickness.

Proof-of-principle measurements were performed with cathodes with grooves of various heights in order to investigate the conformality of thin Boron layers deposited on inclined surfaces by a commonly used coating method like DC magnetron sputtering, as well as the charge collection efficiency as a function of the size of the groove. Also, experimental efforts were directed towards the examination of the merits of the Boron-lined patterned cathode, and the identification of the challenges and limitations of the proposed design. The results of the measurements provided a validation of the model calculations and demonstrated the technical feasibility of the concept. The experimental studies suggest

that cathodes with 45° grooves with a height of 2.1 mm have a good overall efficiency, owing to a rather uniform coating or the sidewalls and efficient collection of the charge generated inside the groove. This design is also in line with the requirement of low material budget in order to reduce the absorption of the incident neutrons in the Aluminium substrate material and also provide a lightweight detector, without compromising its performance.

The next step is to investigate the performance of a detector system consisting of several cathodes with the selected geometry and compare it to the response of a detector incorporating the same number of flat cathodes. Also, a detailed study of the detector operating conditions and tests of various gases is underway in order to further improve the overall performance and establish the design parameters of a complete realistic prototype detector. In parallel, the simulations are being adapted to address the specific issues related to the optimization of a system incorporating several grooved cathodes.

Here we studied both theoretically and experimentally the performance of grooved cathodes coated with  $^{10}\text{B}_4\text{C}$ , but  $^6\text{Li}$  or  $^6\text{LiF}$  could also be used as converter materials. In that case, an optimization of the geometry of the groove for the Li converter is recommended in order to ensure that the alpha particle and the triton generated in the reaction of the incoming neutrons with the converter layer deposited inside the groove have the chance to travel a distance in the counting gas that is large enough to produce a detectable signal.

The ultimate goal of the present studies is to incorporate the selected design in a large-area neutron detector whose performance is competitive to that of a  $^3\text{He}$  proportional counter. The detector will be intended for high-flux neutron scattering applications that require 0.5–1  $\text{cm}^2$  position resolution for thermal neutrons. The preliminary R&D work reported here proves that the proposed design is capable to meet the requirements concerning efficiency per cathode surface area. We also demonstrated that the manufacturing and coating of the cathodes are well within reach of current technologies, which is an important factor in view of building a cost-effective large-area neutron detector.

## Acknowledgments

This research was carried out in the framework of the German In-Kind Contribution to the ESS Design Update, Work Package DE-K2, and supported by the German Federal Ministry of Education and Research (BMBF) under Contract no. 05E10WO1. The authors thank the crew of the Mechanical Workshop of the Physics Department, Technische Universität München, for manufacturing the grooved cathodes.

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